

NAVIGATING CHINA AI MODELS

LLM primer: Critical point of intelligence for global proliferation

From DeepSeek's moment last year (on cost efficiency) to Zhipu's GLM moment this year (on model intelligence)

China's AI open-source/open-weight models are reaching a critical point of intelligence performance vs. global proprietary models, with a significant ramp up in domestic enterprise & global SME adoption that will enable a positive data flywheel of further model improvement, in our view. We evaluate *How* these models achieve such performance at low costs/tight computing resources; *Why* they pursue an open-source/open-weight approach and how they monetize; *What* the key addressable markets are, as global enterprises shift from 'token-maxxing' to ROI-first, where we highlight two favored ARR quadrants for Chinese models; and *Who* the potential long-term winners will be under our Competitive Positioning framework. We note any stricter access to China's future most frontier models for overseas markets, Western markets' policies on China models, and access to high-end computing in model training as three key swing factors/risks to our China AI model token/revenue growth trajectory.

We introduce our Chinese AI model Competitive Positioning framework based on pricing power, cost advantage and financial strength, overlaying with token scale and market share progressions, and identify Knowledge Atlas (Zhipu, initiation) and DeepSeek (private) as the strongest positioned in foundation models, and Bytedance (private) in multi-modal.

Ronald Keung, CFA

+852-2978-0856
ronald.keung@gs.com
Goldman Sachs (Asia) L.L.C.

Eric Sheridan

+1(917)343-8683
eric.sheridan@gs.com
Goldman Sachs & Co. LLC

Allen Chang

+852-2978-2930
allen.k.chang@gs.com
Goldman Sachs (Asia) L.L.C.

Lincoln Kong, CFA

+852-2978-6603
lincoln.kong@gs.com
Goldman Sachs (Asia) L.L.C.

Timothy Zhao

+852-2978-2673
timothy.zhao@gs.com
Goldman Sachs (Asia) L.L.C.

Damian Xie

+852-2978-1398
damian.xie@gs.com
Goldman Sachs (Asia) L.L.C.

Iris Xiao

+852-2978-0202
iris.xiao@gs.com
Goldman Sachs (Asia) L.L.C.

Steve Qiu

+852-2978-2672
steve.qiu@gs.com
Goldman Sachs (Asia) L.L.C.

Luqing Zhou

+852-3465-4207
luqing.zhou@gs.com
Goldman Sachs (Asia) L.L.C.

Eunice Liu

+852-2978-7472
eunice.liu@gs.com
Goldman Sachs (Asia) L.L.C.

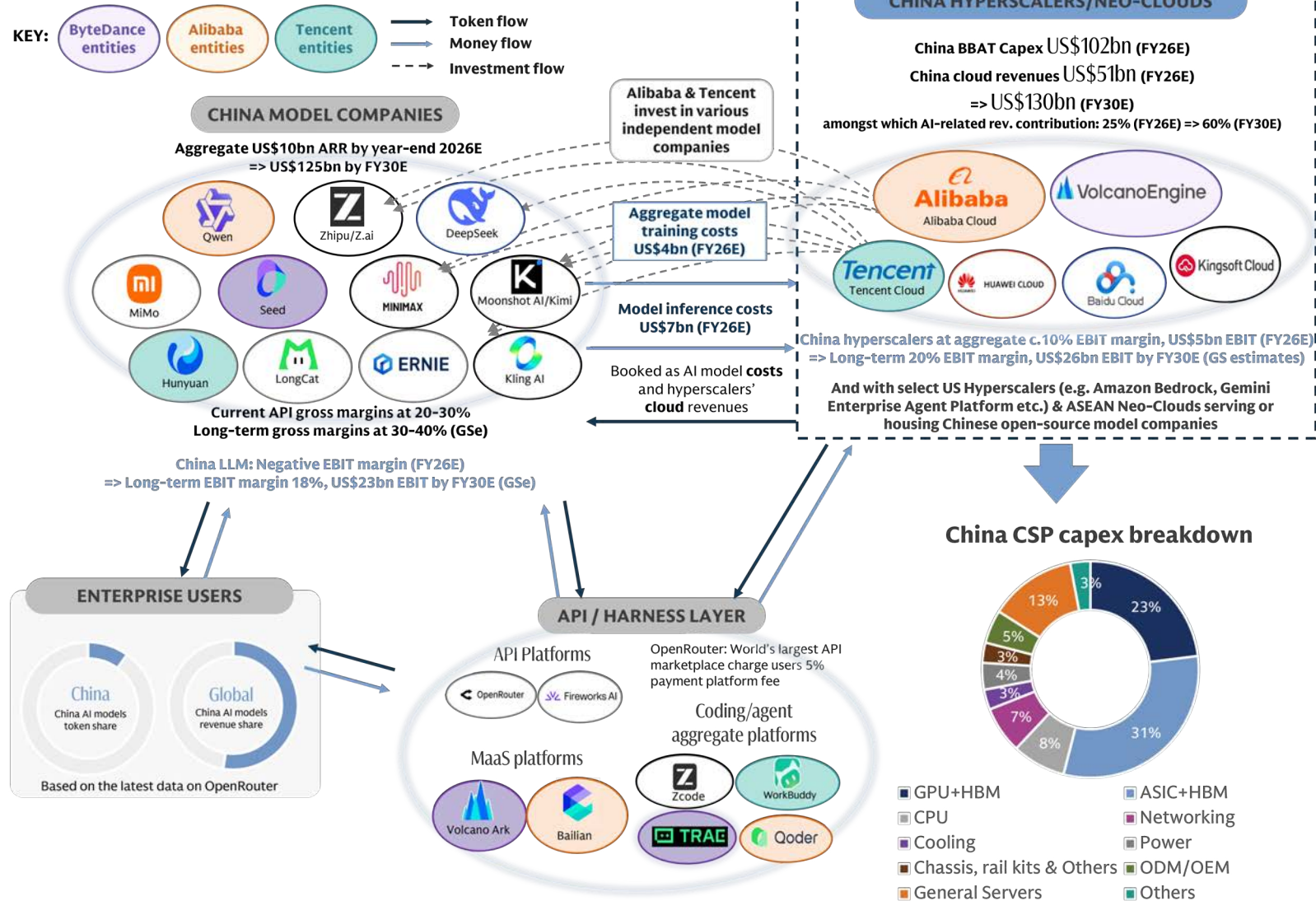
Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

Table of Contents

Dissecting China open-source models unit economics today and path to profitability	4
PM Summary	5
Comparing China's key LLM players	11
Decoding China's AI model tokens & our forecasts on token/revenue share	12
How do Chinese models achieve competitive performance at low costs/tight computing resources?	13
Why are Chinese models pursuing an open source/open weight approach, and ways to monetize?	20
What are the key addressable markets, domestically and internationally, and key risks?	23
Who are best positioned to be the long term winners? Introducing our Competitive Positioning framework	26
Foundation models	26
Multi-modal generation models	32
Key upside and downside risks	37
China's key AI players at a glance	40
Price Target Methodology & Risks	47
Disclosure Appendix	49

CHINA AI MODEL & HYPERSCALERS ECOSYSTEM

Our mapping of key players, revenue/profit pools & interdependencies



1) The calculation of China cloud revenues includes Alibaba, Tencent, ByteDance, Baidu, and Kingsoft Cloud. 2) China CSP capex breakdown is based on GS Greater China Tech team estimates; 3) Companies included are for illustrative purposes and may not be an exhaustive list.

Source: Company data, Goldman Sachs Global Investment Research

Dissecting China open-source models unit economics today and path to profitability

Exhibit 1: Unit economics for China's value-for-money agentic model and top performing coding model, vs. hypothetical economics for global SOTA model**UNIT ECONOMICS OF TWO-TIERED CHINA AI MODELS (PER 1M BLENDED TOKENS)**

	Current (Year-end 26E)			Long-term	
	China's value-for-money agentic model	China's best performing coding model	Hypothetical economics for 'most frontier' SOTA model	China's value-for-money agentic model	China's best performing coding model
API Pricing	\$0.2	\$1	\$3.9	\$0.2	\$2
Inference cost as % of ARR	-0.15 -74%	-0.70 -70%	-0.98 -26%	-0.11 -57%	-1.02 -51%
Gross Profit (ARR - inference cost)	0.03	0.18	2.70	0.07	0.80
% Gross margin	10-15%	15-20%	70%	30-40%	30-40%
Training as % of ARR	-0.06 -30%	-0.40 -40%	-1.89 -49%	-0.03 -13%	-0.25 -13%
EBIT (Gross profit - Training cost & OpEx)	-0.06	-0.39	0.00	0.03	0.44
% EBIT margin	-30%	-39%	0%	14%	22%

CHINA'S VALUE FOR MONEY MODELS BLENDED PRICE: US\$0.2 per blended 1M tokens**CHINA'S TOP PERFORMING CODING MODELS BLENDED PRICE: US\$1 per blended 1M tokens****IN ABSOLUTE US\$ BILLIONS**

	Current (Year-end 26E)			Long-term	
	China's value-for-money agentic model	China's best performing coding model	Hypothetical economics for 'most frontier' SOTA model	China's value-for-money agentic model	China's best performing coding model
ARR, US\$	\$0.8bn	\$1.5bn	\$55bn	\$10.9bn	\$30.6bn
Inference cost as % of ARR	-\$0.6bn -74%	-\$1.1bn -70%	-\$14bn -26%	-\$6.2bn -57%	-\$15.6bn -51%
Gross Profit (ARR - inference cost)	\$0.1bn	\$0.3bn	\$39bn	\$3.6bn	\$12.2bn
% Gross margin	10-15%	15-20%	70%	30-40%	30-40%
Training as % of ARR	-\$0.2bn -30%	-\$0.6bn -40%	-\$27bn -49%	-\$1.6bn -13%	-\$4.4bn -13%
EBIT (Gross profit - Training cost & OpEx)	-\$0.2bn	-\$0.6bn	-\$0.1bn	\$1.5bn	\$6.7bn
% EBIT margin	-30%	-39%	0%	14%	22%

← We estimate aggregate of US\$10bn ARR for China AI model industry by year-end FY26E; we estimate to reach US\$125bn for FY30E

← US\$7bn inference costs for China AI model industry in FY26E, we estimate US\$55bn by FY30E, which translates into hyperscalers'/neo-clouds' revenues

← US\$4bn training costs for China AI model industry in FY26E, US\$20bn by FY30E, as model sizes continue to scale up

← Negative EBIT margin for China AI model companies in FY26E; we estimate to reach 18% EBIT margin/US\$23bn profit pool by FY30E

Source: Company data, Goldman Sachs Global Investment Research

PM Summary

From DeepSeek's moment last year (on cost efficiency) to Zhipu's GLM moment this year (on model intelligence)

China's AI open-source/open-weight models are reaching a critical point of intelligence performance vs. global proprietary models, with a significant ramp up in domestic enterprise adoption and a proliferation of global consumer and SME demand. In this report, we evaluate How these models achieve such performance at low costs/tight computing resources; Why they pursue an open source/open weight approach and how they monetize; What the key addressable markets are, where we highlight two favoured ARR quadrants and risk factors; and Who the potential long-term winners will be under our Competitive Positioning framework.

We believe Chinese models are reaching a critical 'good enough' stage for agentic tasks/specific coding scenarios, and rising fragmentation in China's AI model landscape (but the strong will get stronger). While pricing power remains strong for models with frontier performance/multi-modal, the lower-end segment is in a price war; nevertheless, Agentic AI is driving explosive demand for these value-for-money models at the lower-end. Access to computing will be a swing factor, where US/China regulations, balance sheet and inference efficiency are key.

Accordingly, we introduce our Chinese AI model Competitive Positioning framework based on pricing power, cost advantage and financial strength, overlaying with token scale and market share progressions, and identify Knowledge Atlas (Zhipu, initiation) and DeepSeek (private) as the strongest positioned in foundation models, and Bytedance (private) in multi-modal.

- **How do Chinese models achieve competitive performance at low costs/tight computing resources?** By leveraging smaller-sized parameter models for equivalent benchmark performance, and Mixture-of-Expert and new architectural innovations. Chinese AI models are bifurcating into a two-tiered market where performance and time to market are key to pricing power, thereby leading to two 'ARR maximising' quadrants based on token adoption and pricing. A positive flywheel is taking effect for top Chinese AI models driven by increasing actual real world coding adoption, and reducing their reliance on model distillation practices.
- **Why are Chinese models pursuing an open source/open weight approach, and ways to monetize?** Open source allows for greater flexibility in model training/deployment, and allows for the widest adoption and an open community. We believe open source models' disclosed ARRs are likely understating total deployment and revenue potentials, and we expect more shifts to open weight (with Community License, i.e., commercial terms if for commercial use) among Chinese AI models down the road.
- **What are the key addressable markets, domestically and internationally, and key risks?** We highlight two favoured ARR quadrants and estimate China/China AI models market to see token growth of 25X by 2030E; the coding landscape

to consolidate while the agentic/low-end segment could remain fragmented. International (going global) should be a key upside, with the potential for higher pricing and global proliferation, especially in non-US markets as global enterprises increasingly pivot from a token-maxxing to a ROI-first model that prioritizes clear task boundaries, number of agents per day, back-end process automation and actual output over pure computational token volume. Key risk factors are market access/anti-distillation and regulations, including any stricter access to China's future most frontier models for overseas markets, access to highest-end leased computing equipment (used for training), market access to western markets, further restriction lists/entity list designations (but this could be positive for China's path to further AI self-sufficiency across software/CPU/ASICs), and competition from SLM/threat from AI architectures.

- **Who are best positioned to be the long term winners within China's AI model companies?** We expect players with the largest ARR scale with a gross margin advantage + financial strength to be the long-term winners. We highlight independent AI model companies mostly stand out in our Competitive Positioning framework in pricing power + cost advantage (in aggregate represent over US\$200bn in implied valuations, based on latest market cap/funding rounds), where Zhipu and DeepSeek are the most strongly positioned in text based foundation models, while ByteDance leads in multi-modal capabilities based on our framework.

Signposts we are looking for into 2H 2026

- **Harness/agentic applications:** We expect China AI model companies to increasingly focus on positioning their harness/agentic applications as key entry points, especially in coding (e.g. Zhipu's ZCode, Tencent's Workbuddy, and Alibaba's Qoder which are aggregate platforms that support the full suite of AI models) as model companies attempt to close the loop in capturing more real-life coding and agentic data for scaling. Co-work and industry expert agentic products could be the next priorities. Enterprises will increasingly focus on overall cost per task instead of headline pricing per token, with an openness to using different models (multiple-models approach).
- **Multiple large parameter high-end coding model launches and potentially stricter access to the most advanced Chinese AI models outside of China:** We anticipate multiple new Chinese AI model launches over 2H26 with significantly larger total parameter sizes of 2-5 trillion across Chinese AI model players. Coding/programming segment competition will intensify as Chinese models try to challenge Zhipu GLM's leadership via training on high-quality real-life coding data (where available) and scaling to larger parameter model sizes. We also expect a potential shift from an open-source to an open-weight approach for best-performing models (i.e. from free-for-all use cases to requiring revenue sharing/a take rate for commercial use). The addition of multi-modal/visual understanding will be the next upgrades amongst Chinese AI foundation models (e.g. for GLM, DeepSeek), while MiniMax M3 already excels in these given the multi-modal focus of MiniMax from the start. That said, we note [press reports](#) on potential future restrictions on overseas access to China's most advanced AI models (both closed and open source if at the frontier level) as cutting-edge artificial intelligence is increasingly being seen as a critical national asset.

- **Continued suppressed API pricing for the lower end agentic-focused segment:** While DeepSeek recently announced an increase in peak-hour pricing from mid-July, we expect API pricing and therefore gross margins to remain under pressure at the lower-end pricing segment (around US\$0.1-0.2 per 1M tokens) into the second half, as China's AI model players have significant cash buffers post-fund raising to subsidize competitive pricing at zero/negative gross margins in the near term. As a result, we see financial strength as one of the three most important metrics (alongside pricing power and cost efficiencies) in assessing a Chinese AI model company, where cash on hand, net cash as % of assets and valuation multiples will be the critical financial strength metrics for long-term success. This said, we are Buy-rated on MiniMax as the company stands out on cost efficiency/cost advantage metrics under our Competitive Positioning framework. With its M3 model well positioned in the favored ARR maximizing quadrant (attractive pricing + high token volumes), alongside its discounted valuation at 13X P/2026E year-end ARR (vs. China/global peers which command multiples several times higher at similar ARR stage), we believe risk-reward is skewed to the upside and reiterate our Buy rating. Key swing factors for MiniMax to improve its overall competitive positioning will hinge on its pricing power and financial strength. We expect its H3 video generation model performance (imminent launch in a more favorable industry landscape vs. text models) and time-to-market for its next M3 updates (focusing on further coding intelligence level uplift from post-training/reinforcement learning, we estimate over July-Aug, and a larger parameter size M3 model later in 2H 2026) will be the next key drivers.
- **Multi-modal/video-generation models to see further ARR ramp-up from global adoption:** We expect continued healthy industry pricing and gross margins within video generation (unlike foundation text) where key players ByteDance's SeeDance, Kuaishou's (Buy) Kling and MiniMax's Hailuo/upcoming H3 models to enjoy healthy growth over 2H 2026 amid new functionality breakthroughs (combination of video-generation with LLM) and tight computing resources where demand significantly outpaces capacity. According to China [news reports](#) like LatePost and 36Kr, ByteDance's Seedance gross margins have been at a healthy 70% at its latest US\$2bn+ ARR run-rate.
- We also expect increasing domestic ASICs supply, tighter access to overseas computing resources and potential market access limitations (could mirror TikTok's trajectory where rapid expansion in western markets was followed by more regulations/focus on ensuring data security where computing will have to be conducted within local jurisdictions).

Our assessment of AI model companies: ARR scale x gross margin advantage + financial strength

- Largest ARR scale (Token scale x pricing power)
- Gross margin advantage (Training & Inference efficiency, Technology)
- Financial strength (Balance sheet, Access to computing)

Accordingly, we lay out our Competitive Positioning framework for AI model players based on quantifiable metrics, **1) Pricing Power** (amongst which, based on Time-to-market of model launch, Arena score based on actual usage cases and pricing

level), **2) Cost advantage** (based on token volume scale, throughput/cache hit rate, parameter sizes/activation ratio and our estimate of inference gross margins), and **3) Financial strength** (based on cash on-hand, net cash as % of assets, and valuation multiples).

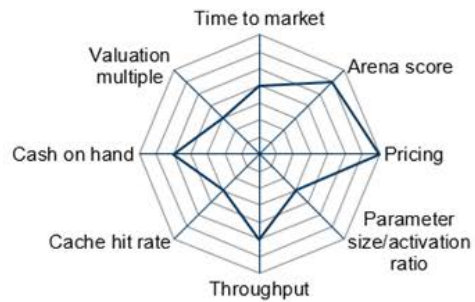
Exhibit 2: Competitive positioning framework for AI model players

Key themes	Chinese models reaching critical stage of intelligence for wide adoption across agentic tasks/specific coding scenarios			Fragmentation between China AI models but strong to get stronger; pricing power to come from frontier performance/multi-modal			Agentic AI driving higher demand for value-for-money models/SLMs			Access to computing, where balance sheet and inference efficiency are key	
Drivers of corporate performance	Competitive positioning										
CP category	Pricing Power			Cost advantage				Financial strength			
CP metrics	Time to market	Arena score	Pricing	Token volume	Throughput /Cache hit rate	Parameter size/ activation ratio	Inference GPM	Cash on hand	Net cash as % of asset	Valuation multiples	

Source: Goldman Sachs Global Investment Research

Exhibit 3: Overview of competitive analysis for key LLM labs' positioning

Alibaba Qwen: Full-stack AI player with multi-year AI/cloud upside, leading position of most areas



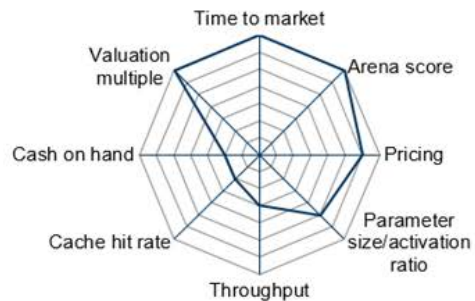
Tencent Hunyuan: Strong contender after AI strategic revamp; potential in Weixin AI agent



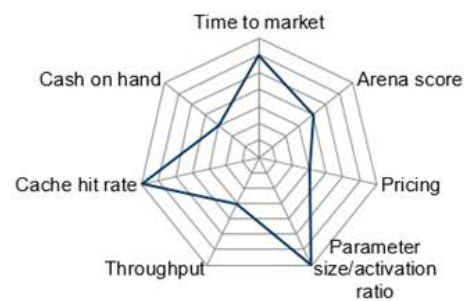
MiniMax M-series: Unique full-modal capabilities among independent players; cost-efficient computing architectures



Zhipu GLM: Leading model capabilities in coding/agentic scenarios



DeepSeek: Extreme cost efficiency to sustain scalable token usage on competitive pricing



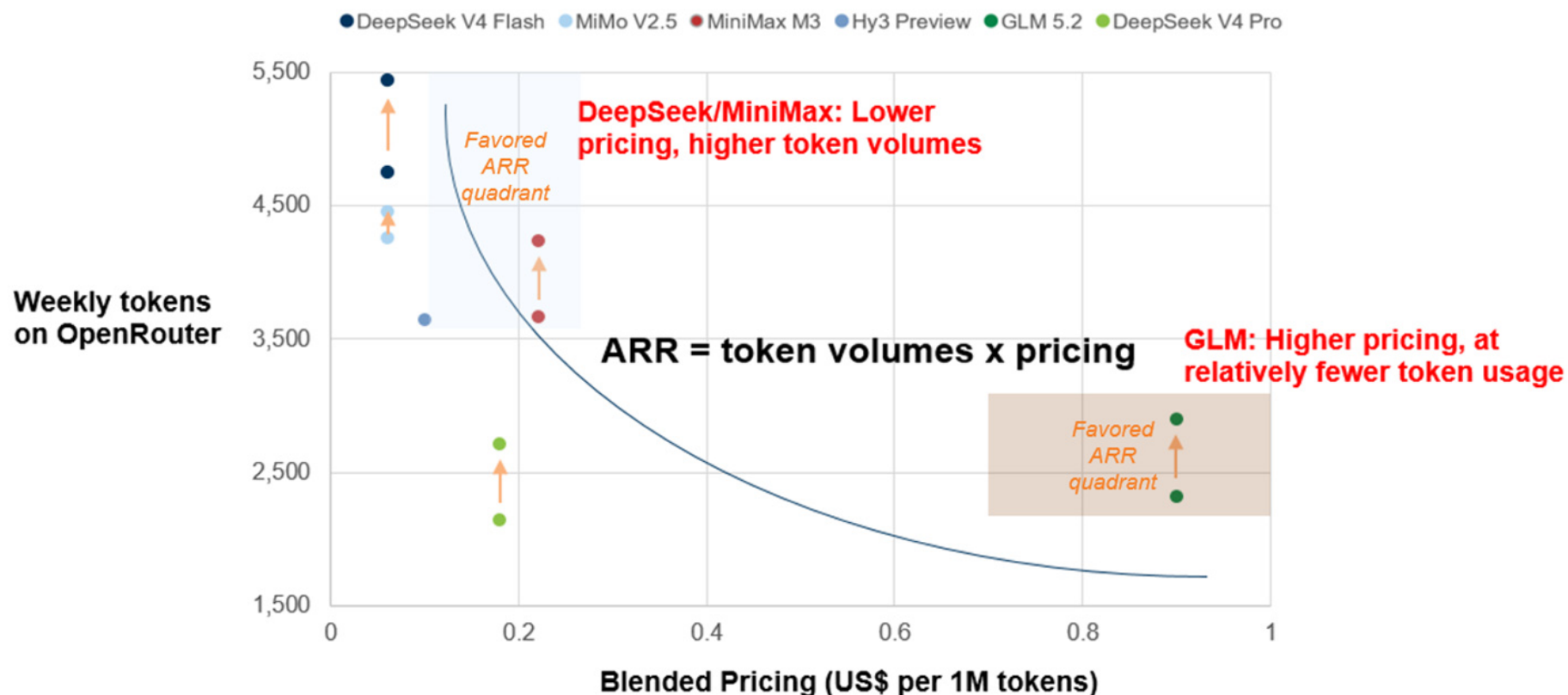
ByteDance Seedance: leading in model performance with strong financial/ecosystem back-up for quick adoption & commercialization



N/A for DeepSeek/ByteDance valuation multiple

Source: Goldman Sachs Global Investment Research

Exhibit 4: We identify two favorable ARR quadrants (in maximizing ARR) which is a combination of maximizing token volumes and/or pricing level



Source: Company data, OpenRouter, Goldman Sachs Global Investment Research

Related research

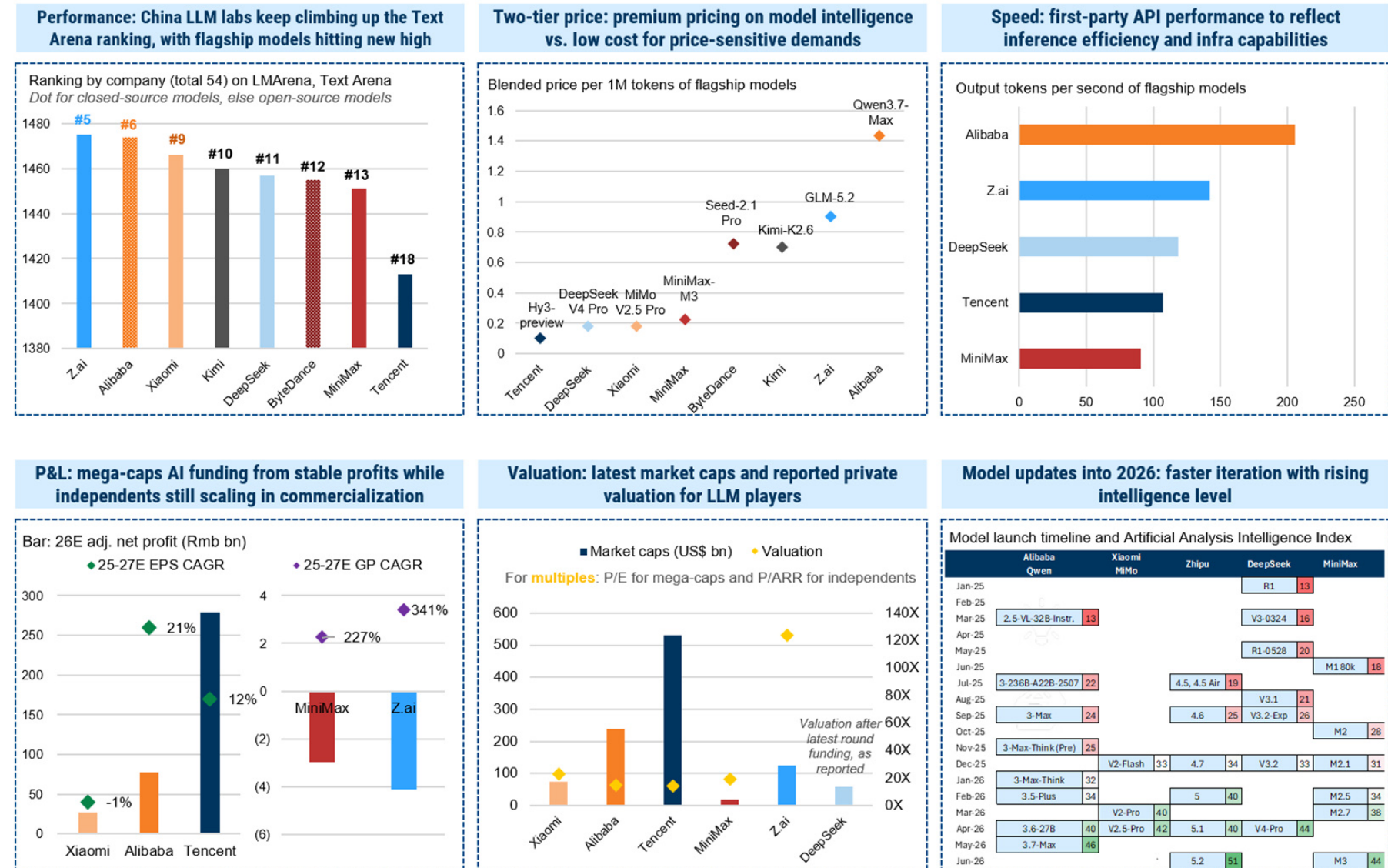
Americas Technology: Decoding the Agentic Economy - The Coming Inflection in AI Usage and Margins, 5 May 2026, by James Schneider, Eric Sheridan and team

Global Server: Raising both AI and General servers; better mix and higher memory costs drive CSP spending, 24 June 2026, by Allen Chang and team

Knowledge Atlas Technology (2513.HK): Initiate with Neutral at US\$110bn valuation; China's top enterprise/coding AI model, July 10, 2026

MiniMax Group (0100.HK): Key focuses post-share unlock; upside skew to risk-reward; Buy, July 10, 2026

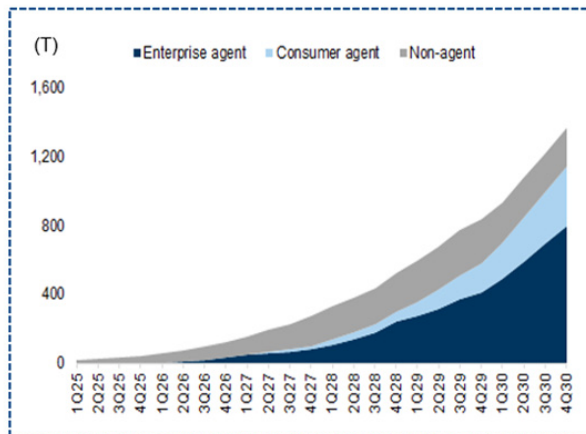
Comparing China's key LLM players



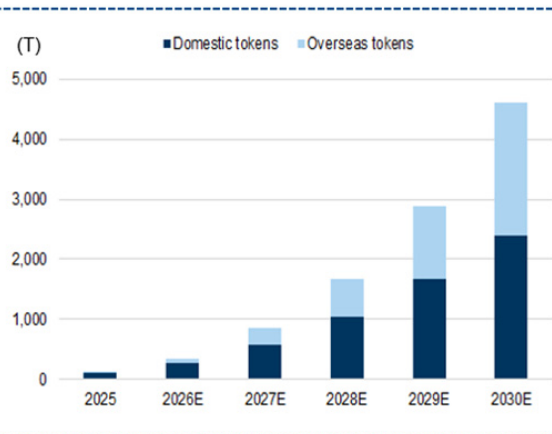
Source: Company data, Goldman Sachs Global Investment Research, WSJ, Artificial Analysis

Decoding China's AI model tokens & our forecasts on token/revenue share

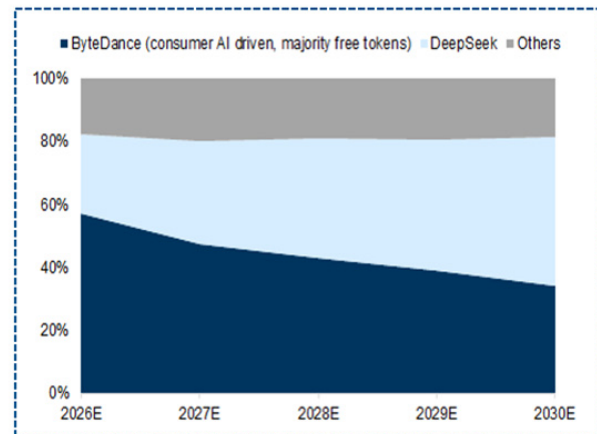
By 2030, we estimate tokens generated by China AI model companies will be 25X higher than today



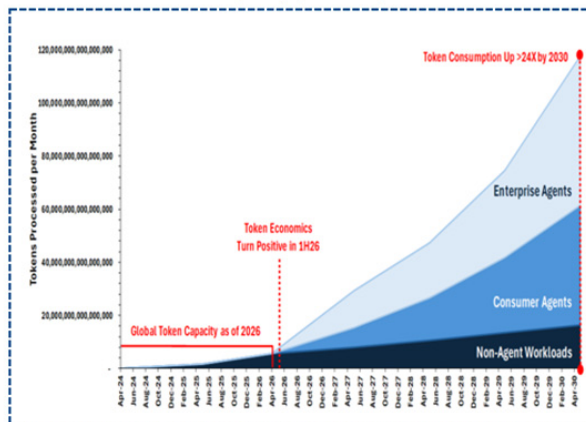
We estimate overseas tokens generated by China AI model companies will reach 55% of global tokens by 2030E



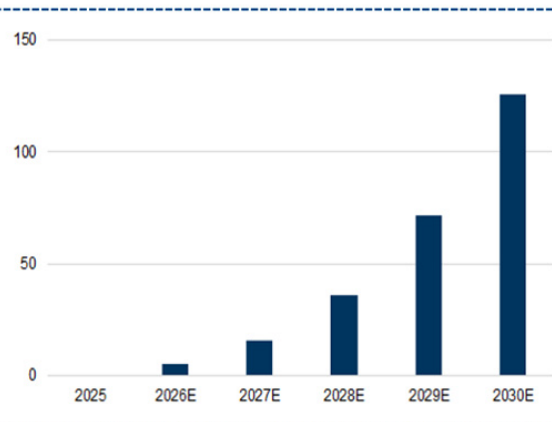
We expect ByteDance and DeepSeek to account for 80% of tokens generated by China AI model companies by 2030E



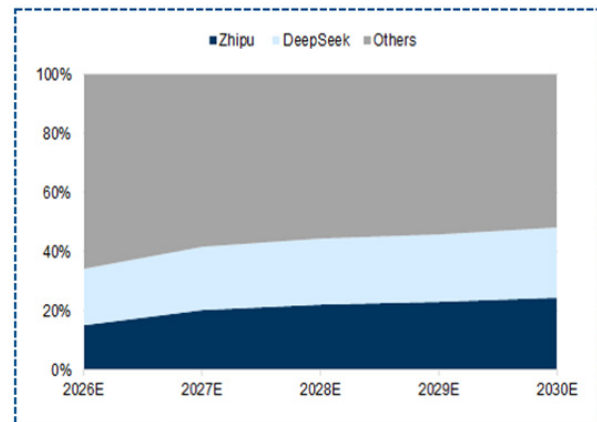
Our US research team estimates consumer and enterprise agents could push token consumption 24X above today's global capacity (ex-China)



We estimate China AI API + subscription revenue pool to reach US\$125bn in 2030E



We expect Zhipu and DeepSeek to capture most of the incremental revenue pool



DeepSeek/ByteDance/Zhipu's (Not Covered) relevant metrics (e.g. token consumption/revenue) are calculated by analyzing the industry and companies that we cover, and then extrapolating them to DeepSeek, ByteDance and Zhipu.

Source: Goldman Sachs Global Investment Research

How do Chinese models achieve competitive performance at low costs/tight computing resources?

Smaller-sized parameters models for equivalent benchmark performances, Mixture-of-Expert and new architectural innovations

Compared with a year ago, Chinese models' coding and agentic capabilities have reached a critical level in being able to complete more coding and autonomous agentic tasks with higher success rates as context windows have been expanded to 1 million tokens. The smaller parameter model sizes of Chinese models (spanning from 200bn to 1.6T parameters, at 2-10% of leading SOTA models, due to constrained access to high-end computing), and highly efficient architectures (MoE, Sparse Attention, OCR etc., at 3-5% activated parameters only vs. total parameter sizes) all contribute to the much lower training and inference costs for Chinese models vs. leading US models. We would attribute the recent step improvement of Chinese models in coding according to Arena.ai to data curation, distillation techniques and reinforcement learning post training, despite their relatively small parameter model sizes (1.6T for DeepSeek V4 Pro, 0.7T for Zhipu's GLM5.2 and 0.4T for MiniMax's M3). On June 27, DeepSeek introduced DSpark, a speculative decoding framework that makes existing DeepSeek-V4 models serve faster. DSpark has already been deployed in DeepSeek-V4 Flash / Pro online serving, improving per-user DeepSeek-V4 generation 60-85% faster on V4-Flash and 57-78% faster on V4 Pro without changing the model's weights or output quality.

Chinese AI models bifurcating into two-tiered market (where performance and time to market are key to pricing power), with two 'ARR maximising' quadrants based on token adoption and pricing

We are seeing pricing power for the highest performing Chinese AI models, e.g. Zhipu's GLM5.2 model and Alibaba's Qwen3.7 Max models at around US\$1 per blended 1M tokens, at 5X that of low-end Chinese AI models. The reported tighter US processes in allowing most SOTA model access has also opened new arenas for China's top performing coding models for enterprises and SMEs. Smaller parameter and activated ratios allow China's top performing models to be priced at US\$1, 10-25% vs. US SOTA models at US\$4-8 per blended 1M tokens, while generating double digit 10-20% gross margins (GSe) that are lower than global SOTA models due to relatively lower pricing power.

In the lower end segment, agentic focused models are priced at US\$0.06-0.2 per blended 1M tokens, which are enabling these models to tap into new global TAMs for price sensitive SMEs and one-man companies. MiniMax generates 60-70% revenues from overseas.

We also note DeepSeek announced that its V4 official version is set for launch in mid-July, alongside the introduction of peak/off-peak API pricing to better allocate resources and improve service stability. V4 Pro/Flash non-peak pricing remains unchanged, while peak hour (9am-12pm/2pm-6pm China time) will be charged at 2X non-peak rates, implying blended

pricing of US\$0.35/US\$0.12 per 1M tokens due to strong Chinese AI model demand that is increasingly causing significant compute tightness in work/productivity scenarios.

Positive flywheel is taking effect for top Chinese AI models from increasing actual real world coding adoption, with less reliance on model distillation ahead

Compared with learning and distillation tactics from global SOTA models in the past, we believe top Chinese models like GLM5 are reaching a critical stage of adoption by China's major enterprises and global users. As per LatePost, AI-generated code has increased to as high as 90% at some China mega-cap companies, up from 20-30% in 2H25, which we believe will enable a positive data flywheel effect of further improvements via reinforcement learning with actual user data (both successful and unsuccessful coding cases) and post training. We think these have underpinned the step improvements in GLM5.2 from GLM5.1 in just over the course of a few months in 2026, and we expect to see further step improvements to Chinese AI models over the next 6-12 months.

Exhibit 5: China's model count is rising in top rank bands across usage (text, coding and agent)

LMarena ranking of Text, WebDev (Coding) and Agent Arena

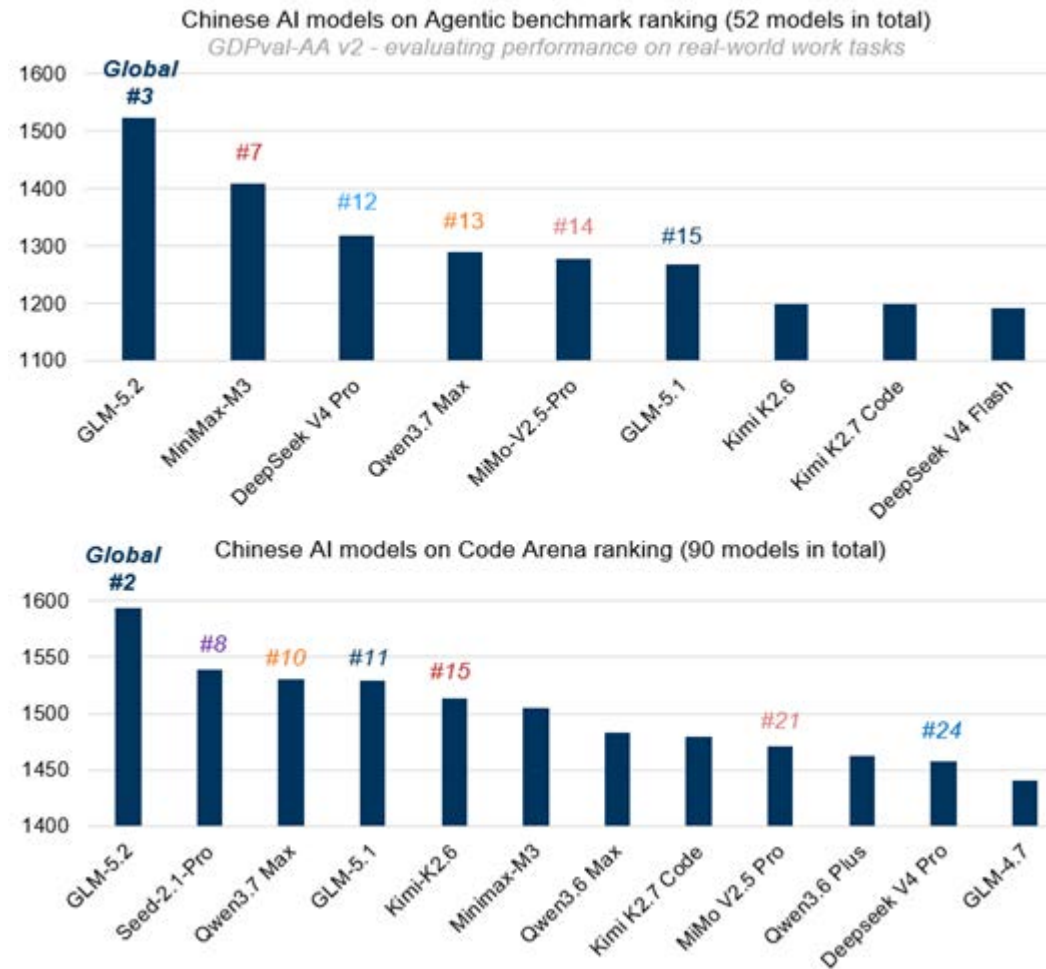
Text Arena			Code Arena			Agent Arena		
	3 months ago	Current		3 months ago	Current		early-June	Current
Top 10% # of China model ranked within the top	7	10 ▲ 3	Top 10% # of China model ranked within the top	0	2 ▲ 2	Top 30% # of China model ranked within the top	0	1 GLM 5.2 at #7
Top 30% # of China model ranked within the top	43	49 ▲ 9	Top 30% # of China model ranked within the top	6	11 ▲ 5	Top 50% # of China model ranked within the top	1	3 ▲ 2
Top 50% # of China model ranked within the top	70	83 ▲ 13	Top 50% # of China model ranked within the top	9	20 ▲ 11	By agentic domain 		
Leading China AI labs 			Best China model rank GLM-5 #9/56 → ZHIPU-AI → GLM-5.2 #2/90			Legal LegalBench Finance Finance Agent v2		

Baseline date for Text/WebDev (Coding) Arena ranking = 3/18/2026, and current date refers to latest date available, i.e. 6/16/2026 & 6/20/2026. Agent Arena ranking started at June 5, 2026, with current date = 6/30/2026

Source: LMArena, Data compiled by Goldman Sachs Global Investment Research

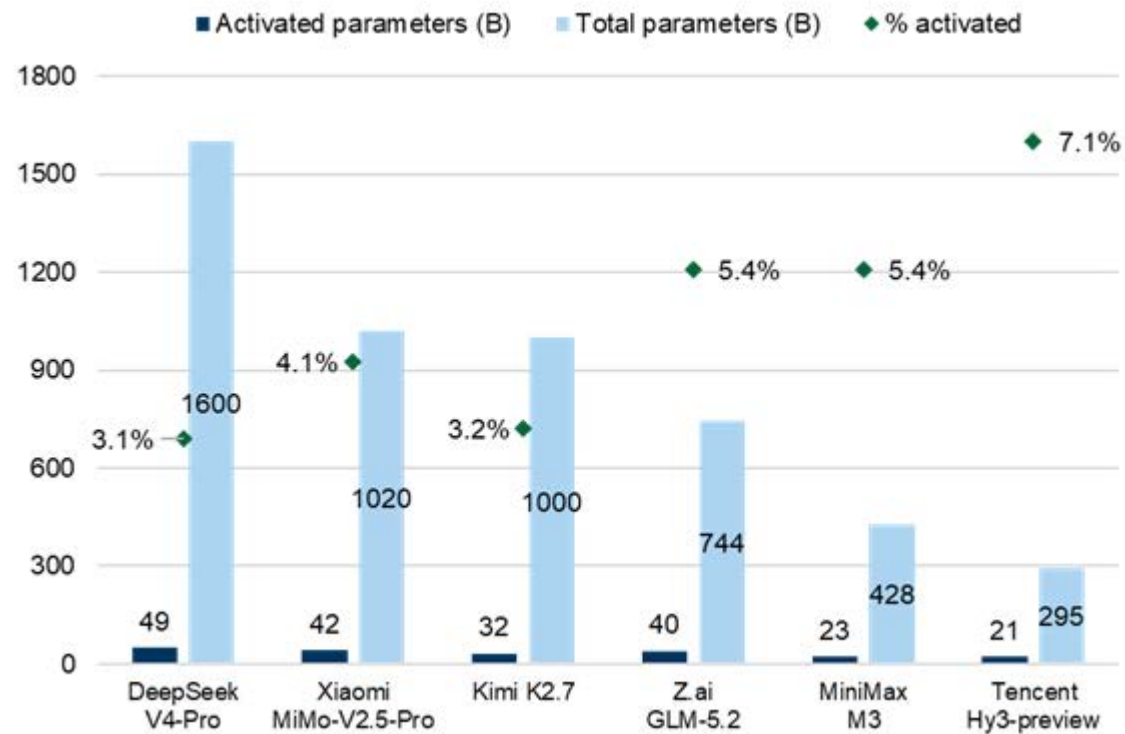
Exhibit 6: Notably in coding/agentic tasks, Chinese players are reaching global top-tier positions

As of Jun 20, 2026



Source: GDPval-AA, LMArena, Data compiled by Goldman Sachs Global Investment Research

Exhibit 7: Compared with global SOTA leading models of several tens of trillions, China open source models are mostly around or below 1 trillion parameter in size, and adopt a MoE structure, with low activated to total parameter ratios for higher inference efficiency



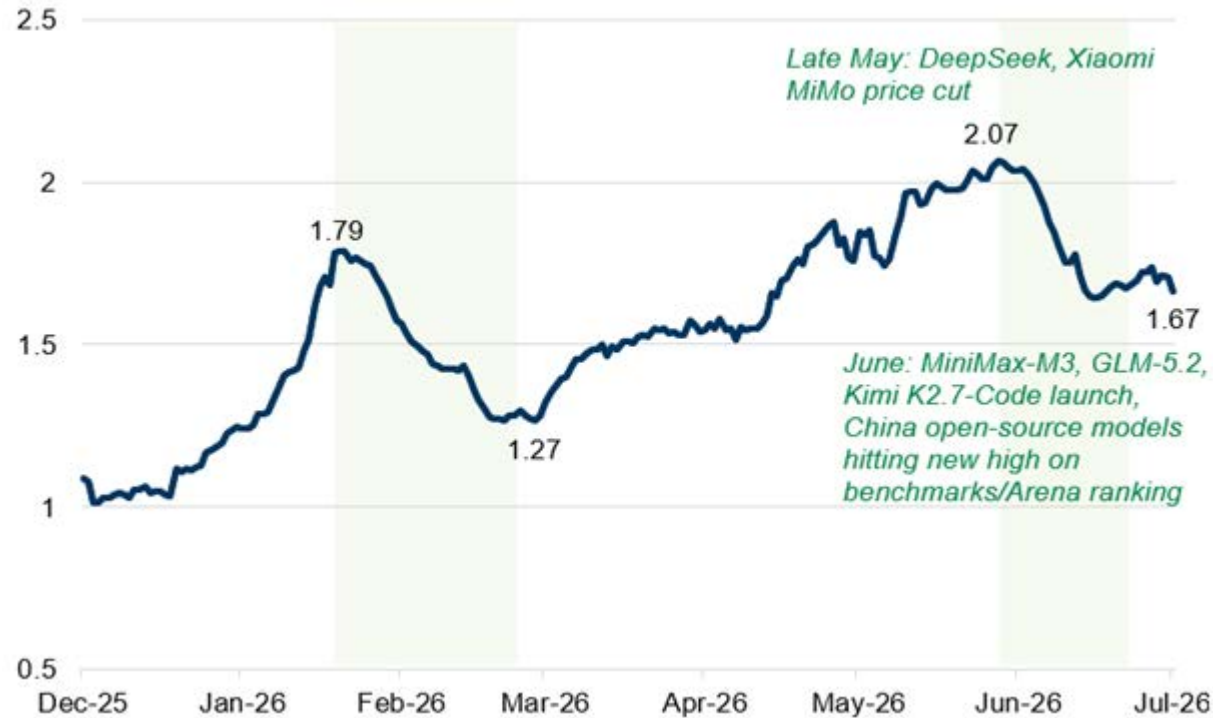
A Mixture-of-Experts (MoE) structure is a sparsely activated neural architecture, where it dynamically selects a small subset of expert networks to process each input, allowing the model to increase its total capacity without activating all parameters during inference.

Source: Company data, Data compiled by Goldman Sachs Global Investment Research

Exhibit 8: Blended LLM token price (SDLLMTK) has been declining since early June, potentially driven by rising adoption of cost effective China models

US\$ per 1M tokens

US\$ per 1M tokens

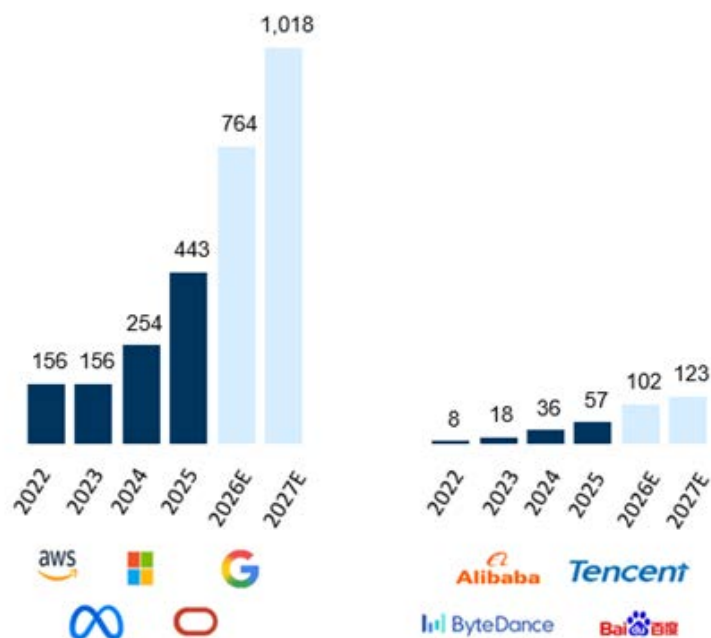


The Silicon Data LLM Token Expenditure Index (SDLLMTK) is a daily blended measure of what the market actually pays to run LLM inference across frontier and open-weight models.

Source: Silicon Data, Bloomberg, Data compiled by Goldman Sachs Global Investment Research

Exhibit 9: Historical and projected capex for major US & China cloud service providers

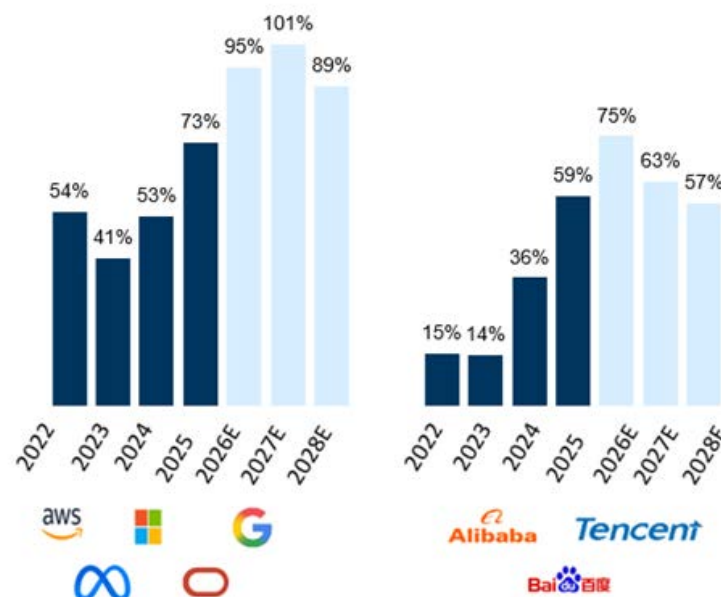
US\$bn



We calculate ByteDance's (Not Covered) relevant operating metrics by analyzing the industry and companies that we cover, and then extrapolating them to ByteDance.

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 10: Capex to operating cash flow ratio is still healthy for China hyperscalers



Source: Company data, Goldman Sachs Global Investment Research

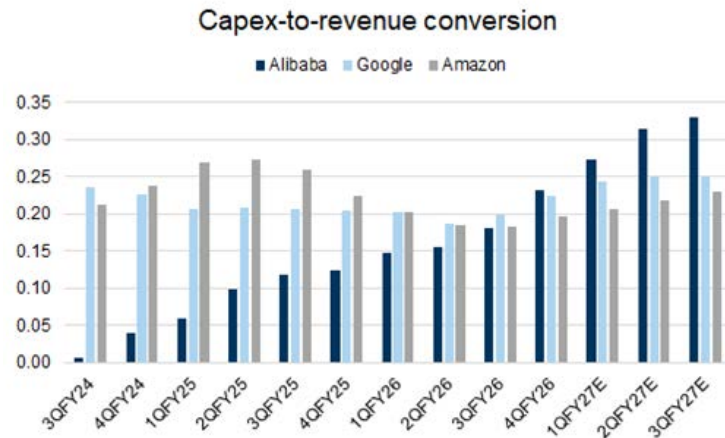
Case study on Meituan's LongCat 2.0: A milestone for China's domestic AI infrastructure

Released on June 30, 2026, we believe Meituan's LongCat 2.0 marks a major milestone as China's first official 1.6 trillion-parameter open-source Mixture-of-Experts (MoE) model trained and deployed entirely on a 50,000-card domestic compute cluster. Purpose-built for agentic coding and complex software engineering workflows, the model features a native 1-million-token context window enabled by LongCat Sparse Attention (LSA) and dynamically activates an average of 48 billion parameters per token to optimize inference costs. **Implications for a more self-sustainable China AI model outlook that is less reliant on foreign high-end chips for model training:** The successful end-to-end pre-training and inference of a trillion-parameter class model on Chinese silicon (reportedly utilizing Huawei Atlas-950 SuperPods) fundamentally drives a more sustainable China's AI model development outlook, in our view. While previous Chinese flagship models, such as DeepSeek V4-pro, mentioned domestic chips for inference, LongCat 2.0's ability to overcome critical memory bottlenecks

and distributed stability challenges during the compute-heavy pre-training phase proves the viability of a wider and more localized hardware stack for AI model training in the future.

Exhibit 11: Alibaba capex-to-revenue conversion ratio vs. Amazon and Google

Rolling 12m incremental cloud revenue/rolling 12m capex (lagged by 1 quarter)

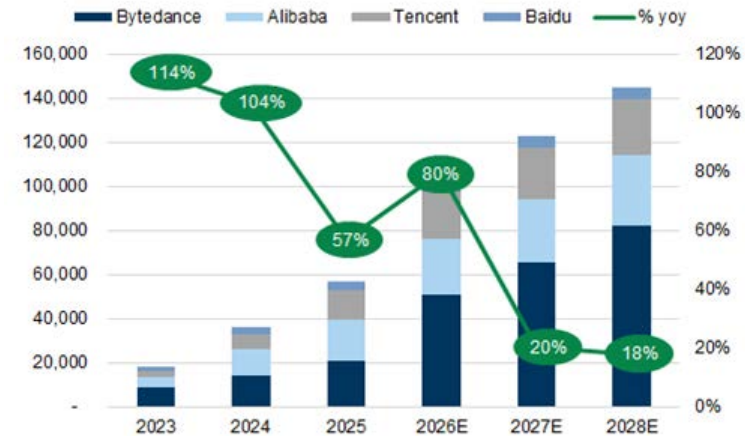


We benchmark Alibaba external cloud revenue growth to Google and Amazon Cloud (lagged c.2 years)

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 12: China Internet yearly capex forecast

US\$m



For ByteDance (Not Covered), we impute capex based on industry trends.

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Why are Chinese models pursuing an open source/open weight approach, and ways to monetize?

Open source allows for higher flexibility in model training/deployment, and allows for the widest adoption and an open community

Alibaba's Qwen model family has long pursued an open source approach (before adopting a closed source for its largest and highest performing Qwen-Max models for better monetization), while other key Chinese AI model players have mostly pursued an open source/open weight approach including DeepSeek, Zhipu's GLM and MiniMax M3 series models with the exception of ByteDance's full closed proprietary approach for its Seed model. We believe the open source approach allows for the highest flexibility in terms of the locations of where models are trained (and thus where the models can be deployed both inside and outside of Mainland China after training). An open source approach also allows for the highest adoption amongst the AI community with full transparency of the model parameters/architecture for trust, and an open community in driving more user feedback and thus model iterations/improvements. The open source approach vs. world's leading closed/proprietary approach also provides an alternative choice for worldwide AI users when the best performing closed models have stricter user access and higher costs from their premium pricing, especially at a time when 'token-maxxing' has become a key cost consideration for many corporates.

Open source models' disclosed ARR are likely understating total deployment and revenue potentials

While open source model companies offer their own coding plans and a chargeable open platform API channel (where the model companies conduct their own model inference), the majority of open source models also allow individuals/third-party hyperscalers/neocloud providers to deploy the models without a charge even for commercial use (e.g. Alibaba Cloud's Bailian MaaS platform can house GLM5.2 open source model without needing to pay a fee/take rate to Zhipu). As a result, while Zhipu's last stated ARR target for year-end 2026 is at US\$1bn, we believe the actual deployment of GLM5.2 model worldwide is and will be multiple-fold higher vs. Zhipu's own API channel token volumes and revenue. There is also increasing post training of Chinese AI models that are re-branded by global companies (e.g. Composer 2 etc.) where the Chinese AI models do not necessarily receive any revenues given the open source spirit.

We expect more shifts to open weight (with Community License) approach among Chinese AI models down the road

While Zhipu's open source GLM model's MIT license allows for free for all use cases (regardless of revenue), MiniMax M series models have pursued a restricted license (where the industry terms it as an open weight with Community License model), which requires MiniMax's agreement and commercial terms (e.g. revenue sharing/a take rate) on commercial use. We believe this will be the likely next path for other open source models in the Chinese AI model industry, in order for eventual gross profits of inference tokens to cover training costs and for each AI model company to achieve a sustainable path to returns.

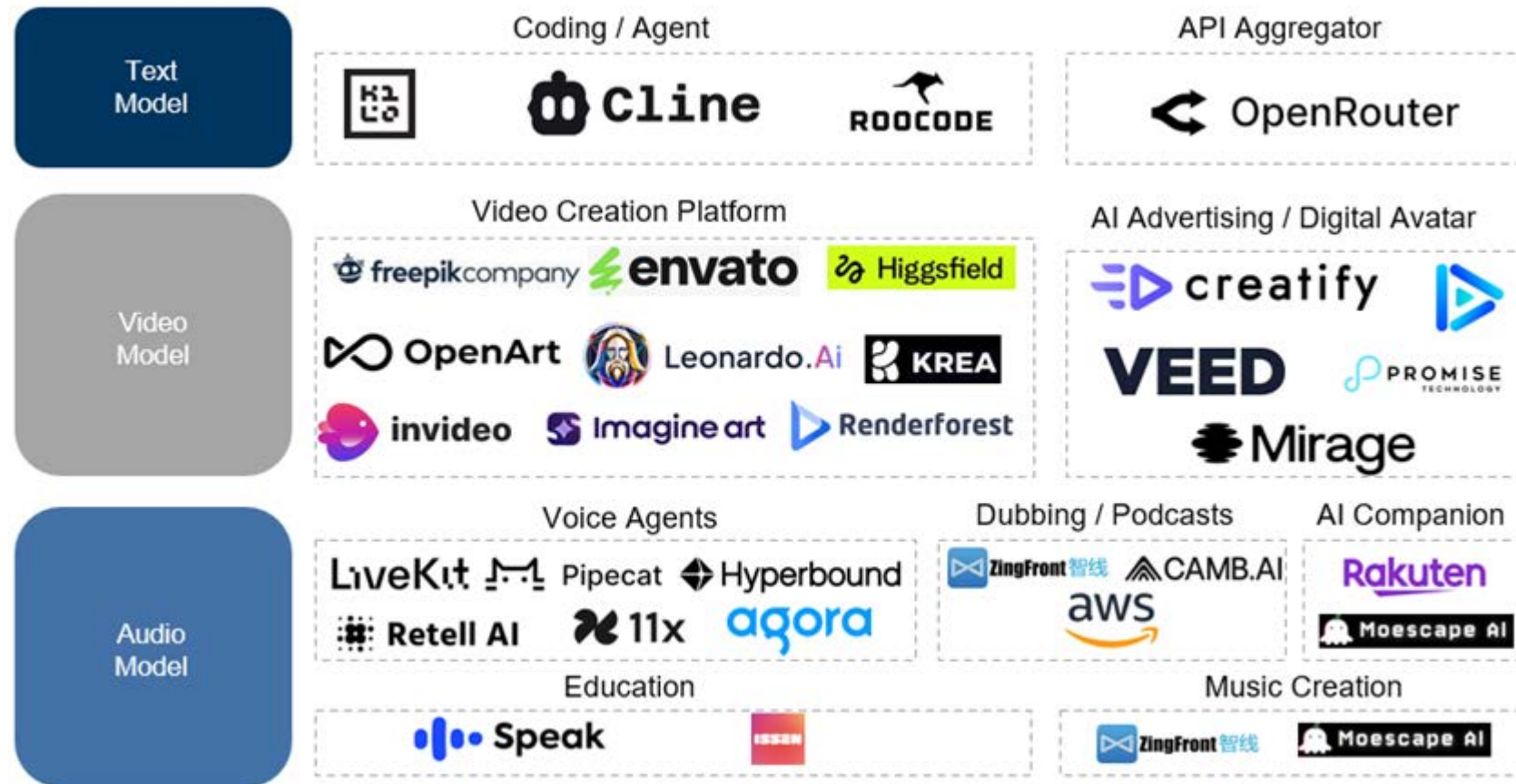
Exhibit 13: Comparison of different open source licenses

Feature	Zhipu GLM 5.2	MiniMax M3
License Type	MIT License (Permissive)	MiniMax Community License (Restricted)
Commercial Use	Unrestricted. Free for all users regardless of revenue	Restricted. Free only for entities with <US\$20M annual revenue
Approval Process	None. No permission needed to deploy commercially	Required. Must email api@minimax.io . <US\$20M revenue needs notice; >\$20M requires formal written license
Attribution	Standard copyright notice retention (MIT terms)	Mandatory "Built with MiniMax" label on products/UI
Prohibitions	Standard legal limitations apply	Explicit restrictions on military use, hate speech, and illegal content

Source: Company data, Data compiled by Goldman Sachs Global Investment Research

What are API platform services?

Chinese AI models typically monetize via subscription plans or their in-house API platform services. API access is typically offered on third party platforms like OpenRouter, Cline or coding agent platforms. As we expect more open source models could shift to open weight, we believe the unit economics of Chinese AI model companies could improve as not only can official API channels be monetized, but also by revenue sharing agreements when these open weight models are deployed and sold on third-party neocloud/hyperscalers like AWS Bedrock, Alibaba Cloud Bailian. Such arrangements could become highly margin accretive as AI model companies will not need to conduct inference/bear computing costs themselves.

Exhibit 14: API platform international enterprise customers through which customers can access Chinese AI models

Source: Company data, Data compiled by Goldman Sachs Global Investment Research

What are the key addressable markets, domestically and internationally, and key risks?

We estimate China AI models' aggregate API+subscription revenue to increase from Rmb35bn in 2026E to Rmb879bn by 2030E from rising model intelligence, in particular with recent models like GLM5.2 reaching a critical point for global adoption and attractive pricing. Our revenue pool estimates for China AI models imply total Chinese AI model daily token consumption of 350T in 2026E to increase to 4,600T by 2030E.

We estimate domestic market to see token growth of 25X by 2030E; coding landscape to consolidate while agentic/low-end segment could remain fragmented

At 140tn daily token volumes for the country in March 2026 and several hundred trillions by June 2026 as per the National Bureau of Statistics, we estimate open source/open weight models have roughly a 30% token share (vs. 70% token market share by ByteDance alone, which is closed source and mainly driven by its Doubao app enterprise and individual user base, as the #1 used AI chatbot in China). We expect similar to the US, the coding segment (at a premium pricing level) will continue to be dominated by SOTA best performing models. Meanwhile, we expect the lower-end segment focused on agentic AI will remain fragmented with multiple players due to the financial strength of AI model companies/mega-caps which would sustain the lower-end segment price war for longer.

International (going global) to be the key upside; with potential for higher pricing and global proliferation, especially in non-US markets

Our US research team estimates ([link](#)) agentic AI will drive 24X growth in token consumption by 2030 (from 2026) to 120 quadrillion tokens per month (or 4 quadrillion tokens per day, from their estimate of 170 trillion daily tokens today), with the biggest driver at 55X from enterprise agents and 12X from consumer agents. The global (ex. China) landscape has seen significant token share gains from Chinese AI models, as rising model intelligence and attractive token costs have driven higher adoption across 24/7 Hermes/Claw/Productivity agents, and shifting global SME mindset on using Chinese models in managing token costs given Chinese models have reached a 'good enough' stage in terms of intelligence/performance.

Pivoting from 'token-maxxing' to ROI-focused metrics, e.g. Daily Active Agents/Agentic Work Units

We believe the AI token proliferation is undergoing a paradigm shift from 'token-maxxing' (an initial focus from late 2025 to early 2026 where enterprises equated high AI token consumption directly with organization productivity) towards an 'ROI-first' model that prioritizes clear task boundaries and output over raw computational volume.

- **'Token-maxxing' has been attributed to corporate inefficiencies and cost overruns:** Data from a Jellyfish AI Engineering trends [study](#) indicated heavy AI users at enterprises consumed 10X more tokens but only had a 2X increase in

output. Meanwhile, we note multiple US Internet companies have commented back in April 2026 that either their engineering teams utilized a full year's AI budget in just four months using agentic products (promoting stricter monthly caps per tool) or changed their internal token utilization leaderboards that previously wrongly incentivized staff to launch inefficient/low value autonomous agent tasks.

- **Enterprise framework is pivoting towards an ROI-first model that prioritizes clear task boundaries, number of agents per day, backend process automation and actual output over pure computational token volume.** Besides a marked increase in adoption of Chinese AI models, global enterprises have been downgrading default models to cheaper flash models for more typical tasks (while reserving SOTA models for only the top most value creating tasks like coding/programming). We see a transition from just token tracking to metrics like Daily Active Agents (DAA) and Agentic Work Units (AWU), and believe the overall cost per task will become more relevant than price per token metrics.

Open source/open weight approach allows for the option for U.S. hyperscalers to host Chinese models, operated within U.S. cloud ecosystem

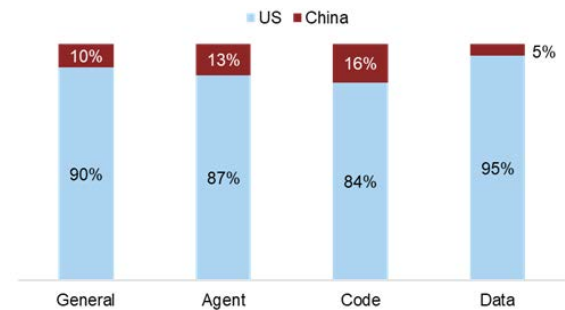
Alphabet and Amazon's (both covered by Eric Sheridan) respective cloud services Gemini Enterprise Agent Platform (via. its Model Garden) and AWS Bedrock already offer a broad selection of Chinese AI models including DeepSeek, MiniMax, Moonshot, GLM and Qwen which are fully managed by the US hyperscalers. Besides the model layer, into applications, worthy of note is Microsoft (covered by Gabriela Borges) CEO's recent remarks at a Wall Street Journal interview ([link](#)) where he noted Microsoft is considering hosting versions of DeepSeek on Copilot as an optional, cost-effective model which could give its customers access to cheaper choices alongside US proprietary models. Microsoft indicated that if it hosts DeepSeek, the model would operate within its cloud ecosystem, ensuring customer data stays inside Azure.

Noting key risks around any potential tighter geopolitical policies given Chinese AI models inroads into western markets.

Key risks to the 'going global' opportunity will hinge on end market access (especially in western countries, and with a focus on where computing is done/data is stored), access to high-end computing for AI model training (that could impact iteration pace and cost structure of Chinese models), and restrictions on Chinese AI model companies could impact supplier relationships/access to US companies and/or access to US capital. More in the key risks section.

Exhibit 15: China AI models now only account for limited share of spend across different tasks, per OpenRouter

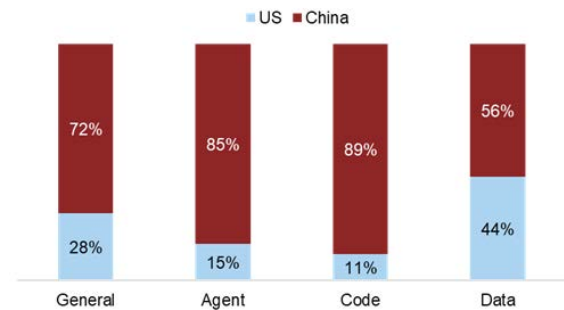
As of week starting Jul 1, 2026



Source: OpenRouter, Data compiled by Goldman Sachs Global Investment Research

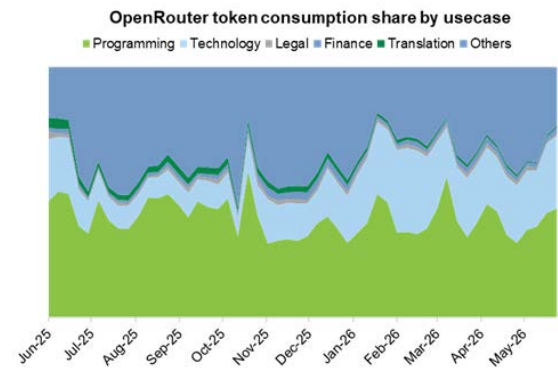
Exhibit 16: While China AI models already capture majority of token usage on price competitiveness, esp. in long duration coding/agentic tasks

As of week starting Jul 1, 2026



Source: OpenRouter, Data compiled by Goldman Sachs Global Investment Research

Exhibit 17: Token consumption is concentrating in complex, high value scenarios like programming/technology



Source: OpenRouter, Data compiled by Goldman Sachs Global Investment Research

Who are best positioned to be the long term winners? Introducing our Competitive Positioning framework

We expect players with the largest ARR scale with a gross margin advantage + financial strength to be the long-term winners.

ARR scale x gross margin advantage + financial strength

- Largest ARR scale (Token scale x pricing power)
- Gross margin advantage (Training & Inference efficiency, Technology)
- Financial strength (Balance sheet, Access to computing)

Accordingly, we introduce our Chinese AI model Competitive Positioning framework based on pricing power, cost advantage and finance strength, overlaying with token scale and market share progressions, and identify Knowledge Atlas (Zhipu initiation) and DeepSeek (private) as the most strongly positioned in foundation models, and Bytedance (private) in multi-modal capabilities.

Foundation models

We assess each key player's competitive positioning from its flagship foundation model, scored across 3 aspects: pricing power, cost advantage and financial strength (of the company), with each aspect further built upon granular quantitative metrics.

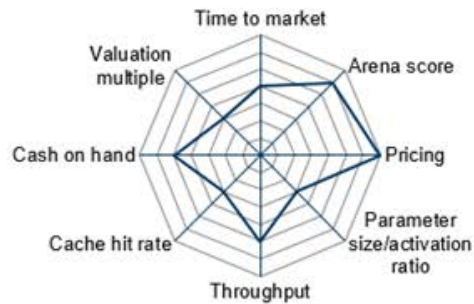
Exhibit 18: Competitive positioning framework for AI model players

Key themes	Chinese models reaching critical stage of intelligence for wide adoption across agentic tasks/specific coding scenarios			Fragmentation between China AI models but strong to get stronger; pricing power to come from frontier performance/multi-modal			Agentic AI driving higher demand for value-for-money models/SLMs			Access to computing, where balance sheet and inference efficiency are key		
Drivers of corporate performance	Competitive positioning											
CP category	Pricing Power			Cost advantage				Financial strength				
CP metrics	Time to market	Arena score	Pricing	Token volume	Throughput /Cache hit rate	Parameter size/ activation ratio	Inference GPM	Cash on hand	Net cash as % of asset	Valuation multiples		

Source: Goldman Sachs Global Investment Research

Exhibit 19: Overview of competitive analysis for key LLM labs' positioning

Alibaba Qwen: Full-stack AI player with multi-year AI/cloud upside, leading position of most areas



Tencent Hunyuan: Strong contender after AI strategic revamp; potential in Weixin AI agent



MiniMax M-series: Unique full-modal capabilities among independent players; cost-efficient computing architectures



Zhipu GLM: Leading model capabilities in coding/agent scenarios



DeepSeek: Extreme cost efficiency to sustain scalable token usage on competitive pricing



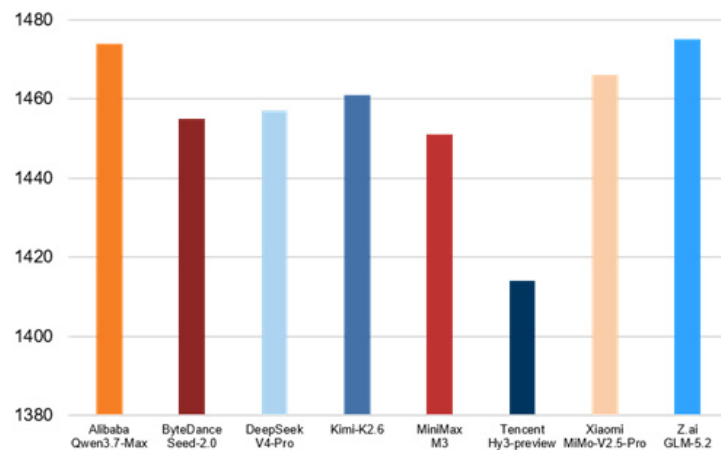
N/A for DeepSeek valuation multiple

Source: Goldman Sachs Global Investment Research

Pricing Power

- **Time to market:** We assess this in terms of how quickly and effectively a player delivers frontier competitive models. From comparing the launch date of the company's flagship models with its prior generation & other models at similar performance level, we aim to assess whether the release delivers a meaningful capability step-up over its past generation, and whether it narrows the gap to the current global SOTA models.

- **Arena score (overall text):** We view model intelligence as the primary determinant of pricing power, since models capable of handling higher-value tasks can deliver clearer ROI and therefore sustain their pricing premium. We use LMArena's score instead of any static benchmark because it reflects a large scale of blind user reviews, making it a more objective read on the real-world capability.
- **Pricing (blended, US\$ per 1M tokens):** We also refer to the realized headline price (across input/output/cached input) for flagship models, where sustained higher pricing with iteration signals stronger pricing power, whereas pricing cuts may indicate a more volume-prioritized strategy.

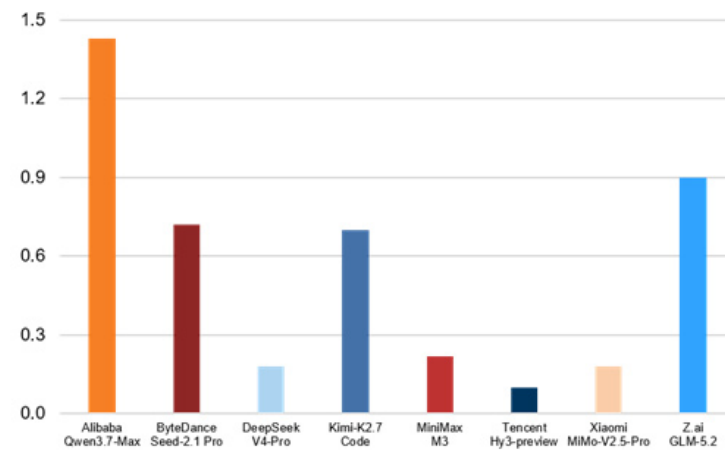
Exhibit 20: Text Arena score of key domestic AI models

As of June 2026

Source: LMArena, Data compiled by Goldman Sachs Global Investment Research

Exhibit 21: Weighted pricing of key player models

US\$ per 1M tokens



As of June 2026

Source: Artificial Analysis, Goldman Sachs Global Investment Research

Cost advantage: the structural cost-to-serve (i.e. inference costs) that sets the floor for price and margin

We refer to the below metrics as proxies for inference efficiency:

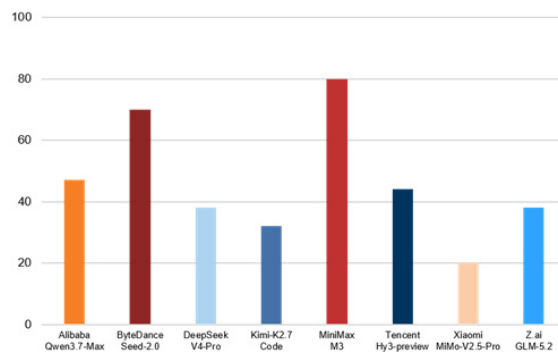
- **Throughput (tokens/second):** Measured by the number of tokens a single GPU generates per second. The more tokens a GPU outputs per second, the more fixed hourly compute cost is spread out. Therefore, throughput is a strong indicator of inference efficiency, which depends on model architecture (sparsity & attention design) and serving efficiency (batching & utilization).
- **Cache hit rate (%):** The share of input tokens served from cache rather than recomputed. In conversations and agentic workflows, much of the input repeats across calls, so models can read those tokens from cache memory instead of

recomputing from new inputs. A higher rate cuts compute, and since cached tokens are near-free to serve despite being billed at a discount (10X-100X cheaper than input cost), it is also margin accretive.

- **Parameter size/activation ratio:** The share of total parameters activated per token (available for open-source MoE models), where fewer active parameters mean fewer FLOPs per token and therefore lower inference cost, at any given level of performance. We also read the absolute parameter size since calculation scales with this degree of freedom.
- **Inference GPM (where disclosed, or GS estimates based on total parameter size/activated parameters):** The realized gross margin on model API, where ~90% of COGS is inference costs, as a direct indication on cost efficiency.

Exhibit 22: Throughput of key player models (tokens per second)

Processing capacity per unit time, representing how fast the model writes

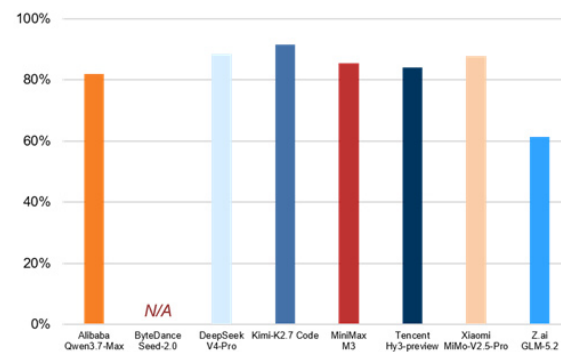


As of June 2026

Source: OpenRouter, Data compiled by Goldman Sachs Global Investment Research

Exhibit 23: Cache hit rate of key player models

% of request served from cache memory instead of new inputs, higher = better for cost/speed

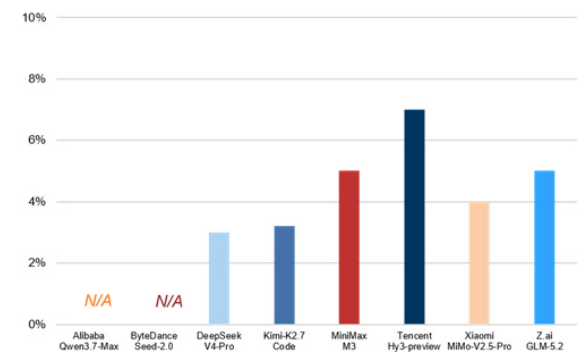


As of June 2026

Source: OpenRouter, Data compiled by Goldman Sachs Global Investment Research

Exhibit 24: Active to total parameter ratio of key player models

Share of parameters activated per token; lower means cheaper inference costs given the same performance



Parameter data unavailable for closed-source models (Alibaba's Qwen and ByteDance's Seed); As of June 2026

Source: Company data, Data compiled by Goldman Sachs Global Investment Research

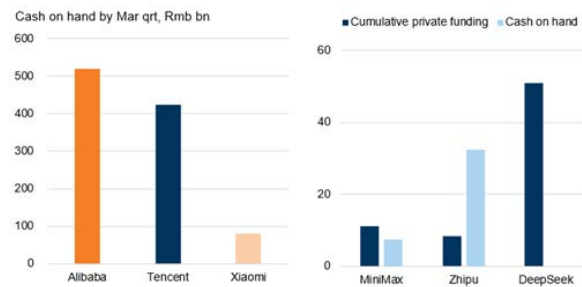
Financial strength: capacity to keep funding frontier R&D and training before a profitability turnaround

Cash on hand and net cash/debt as % of assets: We use total cash on hand as a measure of balance sheet strength, with net cash as % of assets to normalize across players of different scale. We see mega-caps (compared with individual players) as having greater resources to accumulate compute, fund multi-modal exploration and push faster product distribution.

Valuation multiples: For independents, we use P/ARR 2026E multiples as they are not yet profitable and P/ARR is a more comparable measure of business scale and future monetization potential, and for mega-caps we use P/E 2026E.

Exhibit 25: Cash on hand for key players

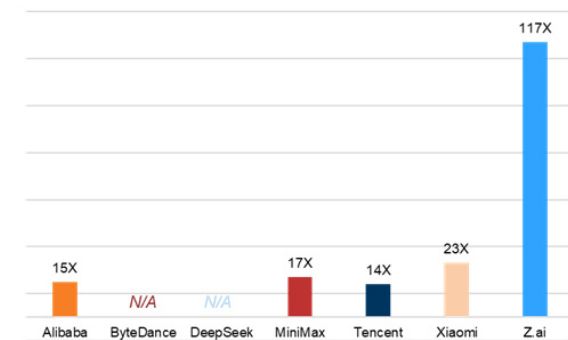
Rmb bn



Source: Company data, The Information, Data compiled by Goldman Sachs Global Investment Research

Exhibit 28: Valuation multiples for key players

Mega-caps valuation on P/E, independent players valuation on P/ARR, 2026E

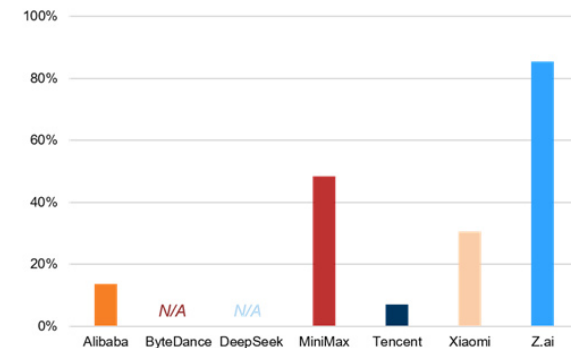


P/E for Alibaba, Tencent, and Xiaomi; P/ARR for MiniMax and Z.ai; Data as of Jun 26, 2026

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 26: Net cash as % of total asset for key players

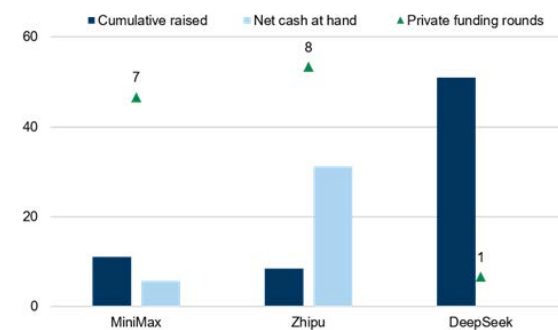
As of July 2026



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 29: Independent AI players' cumulative external private funding (actual) and GSe for net cash as of end-2026E

Rmb bn

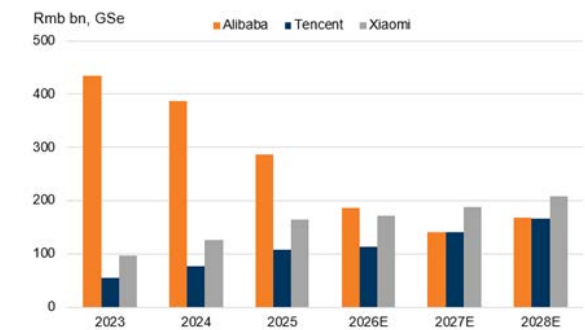


N/A for DeepSeek's net cash at hand

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 27: Net cash estimates for Alibaba, Tencent and Xiaomi

Rmb bn, GSe

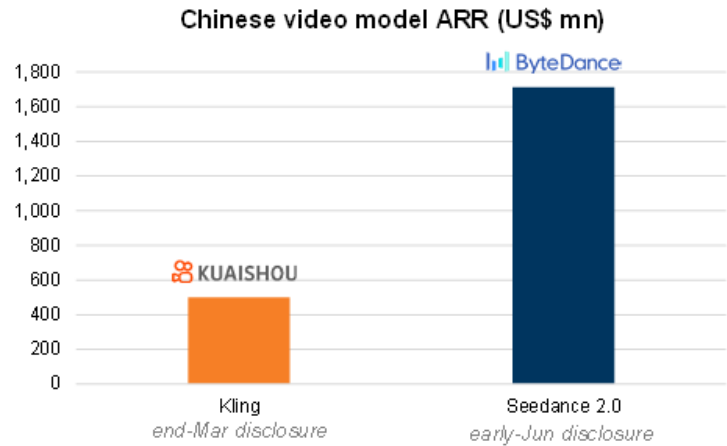


Source: Company data, Goldman Sachs Global Investment Research

Multi-modal generation models

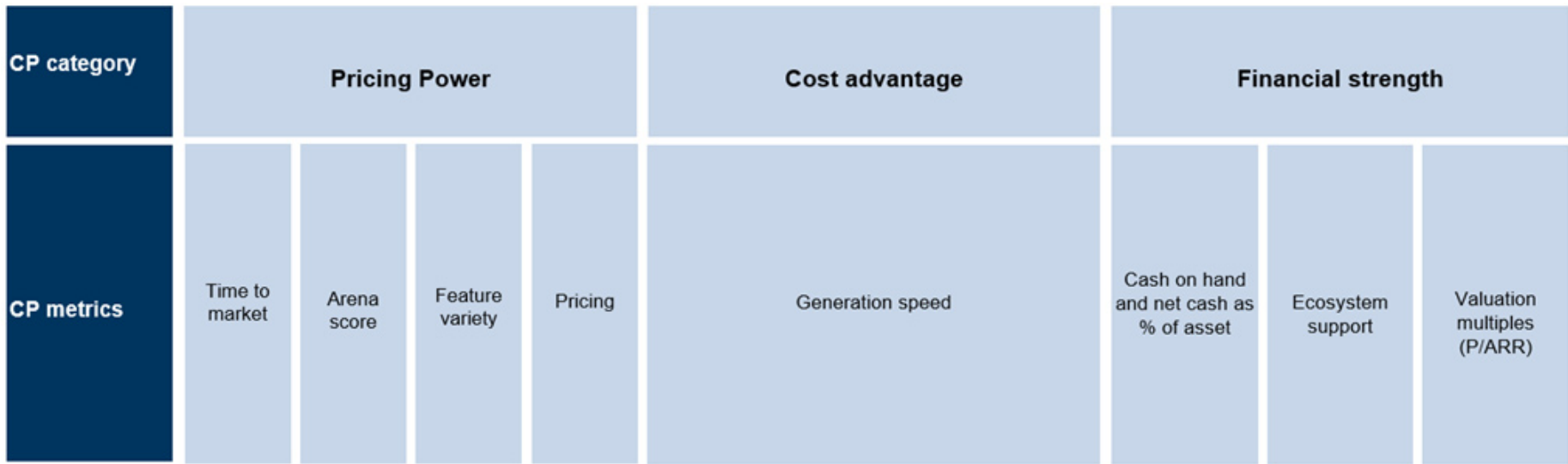
We assess the competitive positioning of video generation players (ByteDance, Alibaba, Kuaishou and MiniMax) under a similar framework across 3 dimensions: pricing power, cost advantage, and financial strength, with each built upon more detailed metrics.

Exhibit 30: Latest disclosure of Chinese AI video model ARR (US\$ mn)



Source: Company data, 36kr, Data compiled by Goldman Sachs Global Investment Research

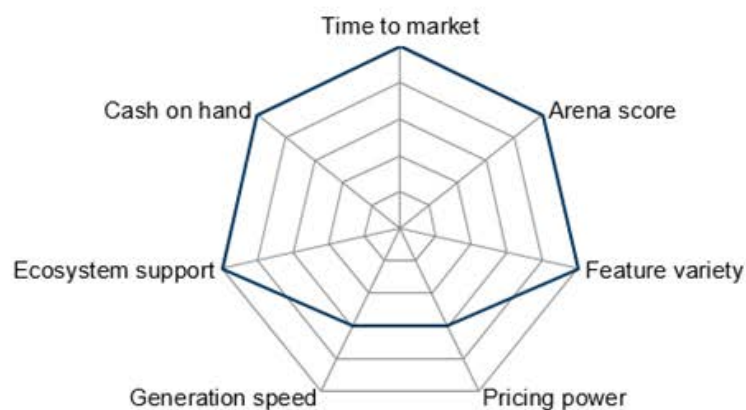
Exhibit 31: Competitive positioning framework for multi-modal generation players



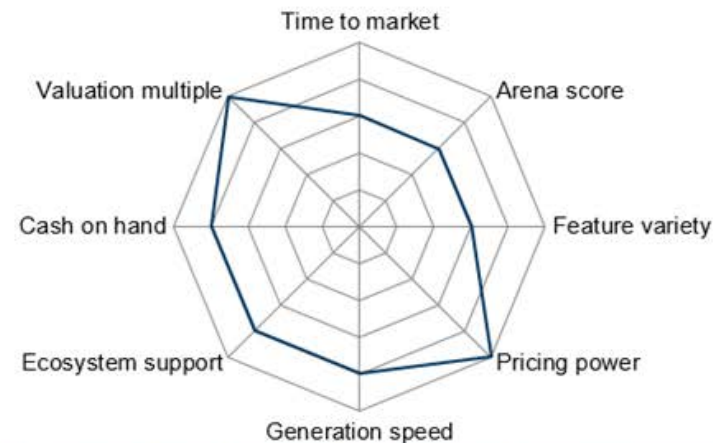
Source: Goldman Sachs Global Investment Research

Exhibit 32: Overview of competitive analysis for key multi-modal labs' positioning

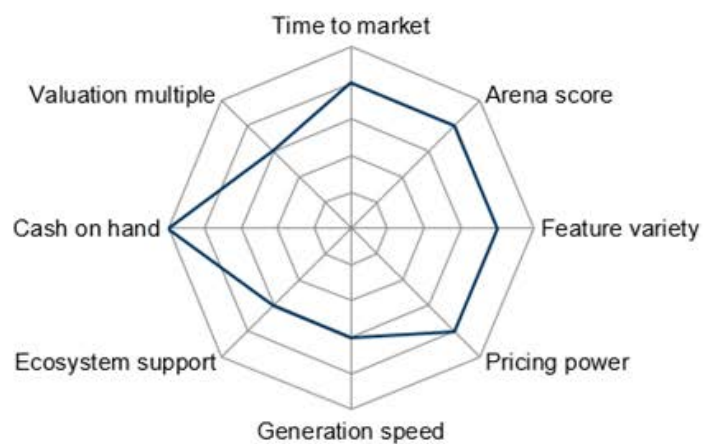
ByteDance: leading in model performance with strong financial/ecosystem back-up for quick adoption & commercialization



Kuaishou: prioritizing enterprise & professional users with strong pricing power and clearer monetization path, supporting a more manageable ROI



Alibaba: abundant resources to support rapid iteration for catching up in performance / feature offerings



MiniMax: native multi-modal training (text+video) to offer synergy on cost efficiency & capability, with anticipated H3 launch as key catalyst



ByteDance's (Not Covered) relevant metrics are calculated by analyzing the industry and companies that we cover, and then extrapolating them to ByteDance. N/A for ByteDance's valuation multiple

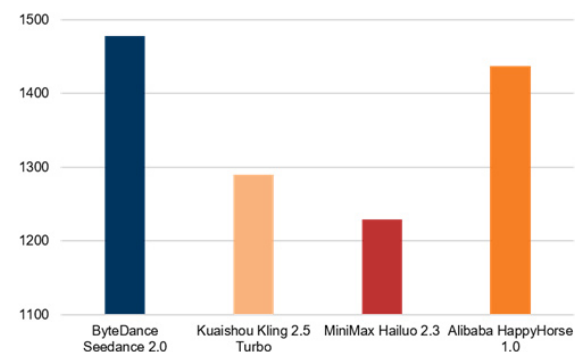
Source: Goldman Sachs Global Investment Research

Pricing Power

- **Time to market:** Measured by how quickly and effectively a player delivers frontier competitive models. From comparing the launch date of the company's flagship models with its prior generation & other models at a similar performance level, we aim to assess whether the release delivers a meaningful capability step-up over its past generation, and whether it narrows the gap to the current global SOTA models.
- **Arena score (average from text-to-video generation and image-to-video generation):** We view model intelligence as the primary determinant of pricing power, since stronger generation models handle higher-value tasks/commercial work and deliver clearer ROI. We use the LMArena score instead of any static benchmark because it reflects a large scale of blind user reviews, making it a more objective read on the real-world capability.
- **Feature variety:** Reflects the generation quality and accessibility of the model, where we further look at **1) native audio (Y/N):** whether the model directly outputs audio matched to the visuals, rather than requiring a separate voice/audio model layered on; **2) max duration:** how long a single clip the model can generate; **3) max # of storyboards:** how many distinct shots the model can generate within one clip, implying its ability to sustain narrative coherence; and **4) max # of multi-modal reference inputs:** how many reference materials beyond text prompt (images or video clips) the user can feed in to guide generation.
- **Pricing (US\$ per min of 1080p video from text-to-video generation, without audio):** We also refer to the realized headline price for flagship models, where sustained higher pricing with iteration signals stronger pricing power.

Exhibit 33: Video Arena score of key players

Average score of text-to-video generation and image-to-video generation

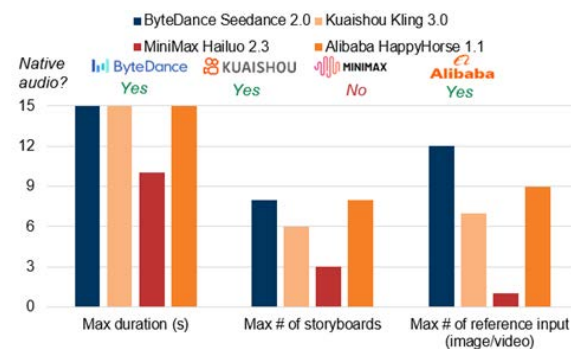


As of Jun 2026

Source: LMArena, Data compiled by Goldman Sachs Global Investment Research

Exhibit 34: Feature variety of key players

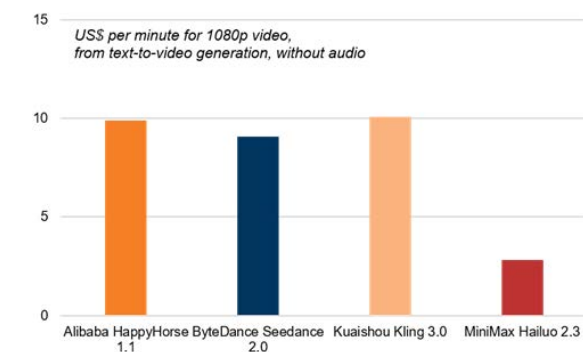
Native audio (Y/N), max duration (s), max # of storyboards and max # of reference inputs (as image/video, beyond text prompt)



Source: Company data, Data compiled by Goldman Sachs Global Investment Research

Exhibit 35: API pricing of key players at the same settings

US\$ per minute



ByteDance Seedance 2.0 datapoint is for 720p video generation (as default setting), all else equal, as per Artificial Analysis

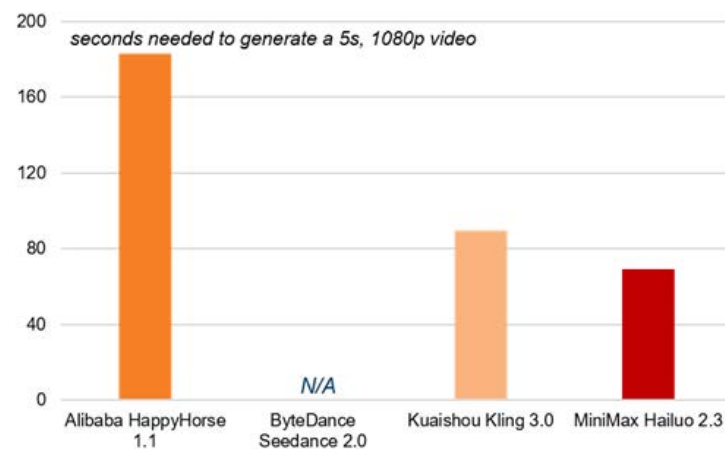
Source: Artificial Analysis, Data compiled by Goldman Sachs Global Investment Research

Cost advantage

- **Generation speed:** Seconds to generate a 5s 1080p clip, where faster speeds indicate higher unit compute utilization and lower inference cost per video.

Exhibit 36: Generation speed of key players

seconds, lower = better



As of early-July, 2026

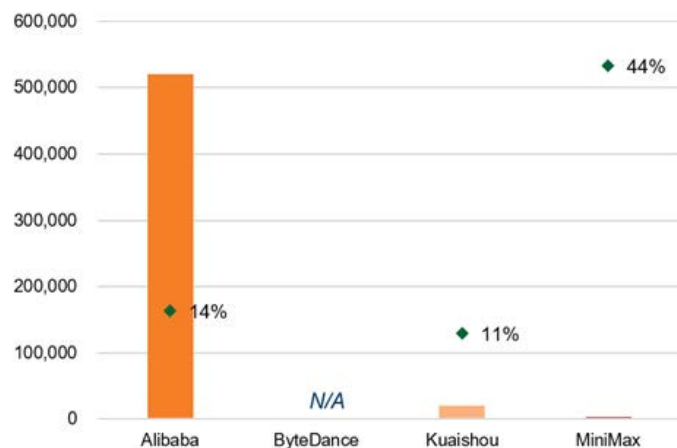
Source: Artificial Analysis, Data compiled by Goldman Sachs Global Investment Research

Financial strength

- **Cash on hand and net cash/debt as % of assets:** We use total cash on hand as a measure of balance sheet strength, with net cash as % of assets to normalize across players of different scale. We see mega-caps (compared with individual players) as having more resources to accumulate compute, fund multi-modal exploration and push faster product distribution.
- **Ecosystem support:** Customer acquisition and distribution channels within the player's ecosystem (including content platforms, editing tools and MaaS), where platform companies such as ByteDance and Kuaishou have an edge in strengthening enterprise commercial adoption and consumer traffic acquisition, thereby powering their data flywheels.
- **Valuation multiples:** For both MiniMax and Kuaishou (with Kling AI valuation at US\$18bn post-money implied valuation, see our team's [analysis](#)), we use P/ARR 2026E multiple.

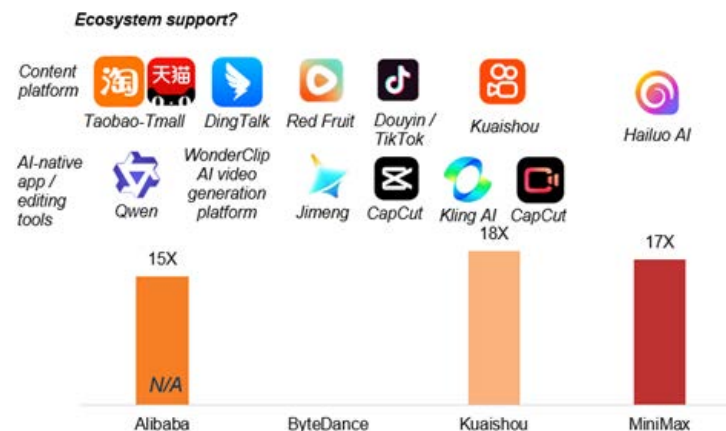
Exhibit 37: Cash on hand and net cash as % of total assets by 2025 for key players

Rmb mn



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 38: Ecosystem support and valuation multiples of key players
P/ARR multiples, as of end-2026E



Source: LatePost, Company data, Goldman Sachs Global Investment Research

Key upside and downside risks

Upside risks:

- **Stronger-than-expected model intelligence uplift from self-sustainable positive flywheel:** Breakthroughs in coding and multi-modal performances, offering seamless integration and understanding across text, image, and audio, would unlock new applications, physical AI and AGI potentials.
- **Faster-than-expected path to profit visibility:** If Chinese AI model companies shift faster toward being open-weight (from pure open source) and demonstrate superior operational efficiency, such as optimizing model inference costs and training infrastructure, or diversify monetization channels to achieve higher revenue growth or better unit economics sooner, it would accelerate the timeline to positive cash flow and net income. Early profitability would boost investor confidence on the AI model industry, potentially reducing the need for future external funding.
- **Additional non-inference take rate revenue streams:** If Chinese AI model companies are able to train and achieve SOTA model performance, an open-weight approach charging take rates/revenue sharing on neoclouds/hyperscalers that house such models and sell to customers may transform AI model company business models to a high gross margin business (without needing to conduct inference for each token sold, but earning via revenue sharing with platforms that

sell top performing Chinese models).

- **Stronger-than-expected commercialization capability:** Growth of new applications, Code and Co-Worker platforms, coupled with ‘digital worker’ agent models could provide multiple channels for revenue generation. If these channels achieve higher-than-expected adoption rates, conversion efficiencies, or ARPU, Chinese AI models could outperform current revenue forecasts.

Downside risks:

- **Competition in global foundation model industry, vs. US/China mega-caps:** Chinese AI models operate within the intensely competitive and rapidly evolving global foundation model industry, where the risks include the challenge of keeping pace with swift technological advancements, frequent new product introductions, and dynamic customer demand. Failure to adapt products and invest significantly in R&D in a cost-effective and timely way, could lead to diminished competitiveness. In addition, the early-stage nature of the market opportunities makes customer demand unpredictable and an inability to meet user requirements would risk losing existing users and adversely impact revenue.
- **Limited near-term profit visibility due to high R&D expenses:** Achieving and sustaining profitability of Chinese AI model companies in the near future could be a challenge, as success depends on continuous technological development, user experience improvement, and effective commercialization. Failure to manage escalating costs and generate adequate revenue could materially impact the company’s financial position and profitability.
- **Risks around IP/content generation:** Chinese AI models, especially multi-modal generation models, face risks from the generation of inappropriate content. Similar to challenges encountered by other AI applications, such as lawsuits against Character AI’s chatbot for its alleged role in teens’ safety, image/video-generation models could be exposed to reputational damage and potential lawsuits. Robust content moderation and ethical AI development are crucial to prevent adverse output and ensure a safe user experience. Chinese companies could also face risks from third party intellectual property/distillation method infringement claims, from both the data used to train their models and the content they generate.
- **Cash burn/self-funding capability:** The intense capital requirements for advanced AI model training, coupled with rapid technological advancements by competitors, could pose financing risks to the relatively fragmented Chinese AI model landscape. In addition, any models that fail to keep pace with leading peers could run into difficulties in generating sufficient revenue (ARR x inference gross margins) to cover training costs.
- **Risk around geopolitics under an intensified tech race between the US and China:** Any US restrictions on the use of Chinese AI models could limit market access and adoption, and Chinese AI model companies could also face increased costs if barred from training its models on US cloud service providers or be included into any list which limits procurement of US supplies from hardware to software. The evolving global chip availability dynamics also present a critical supply

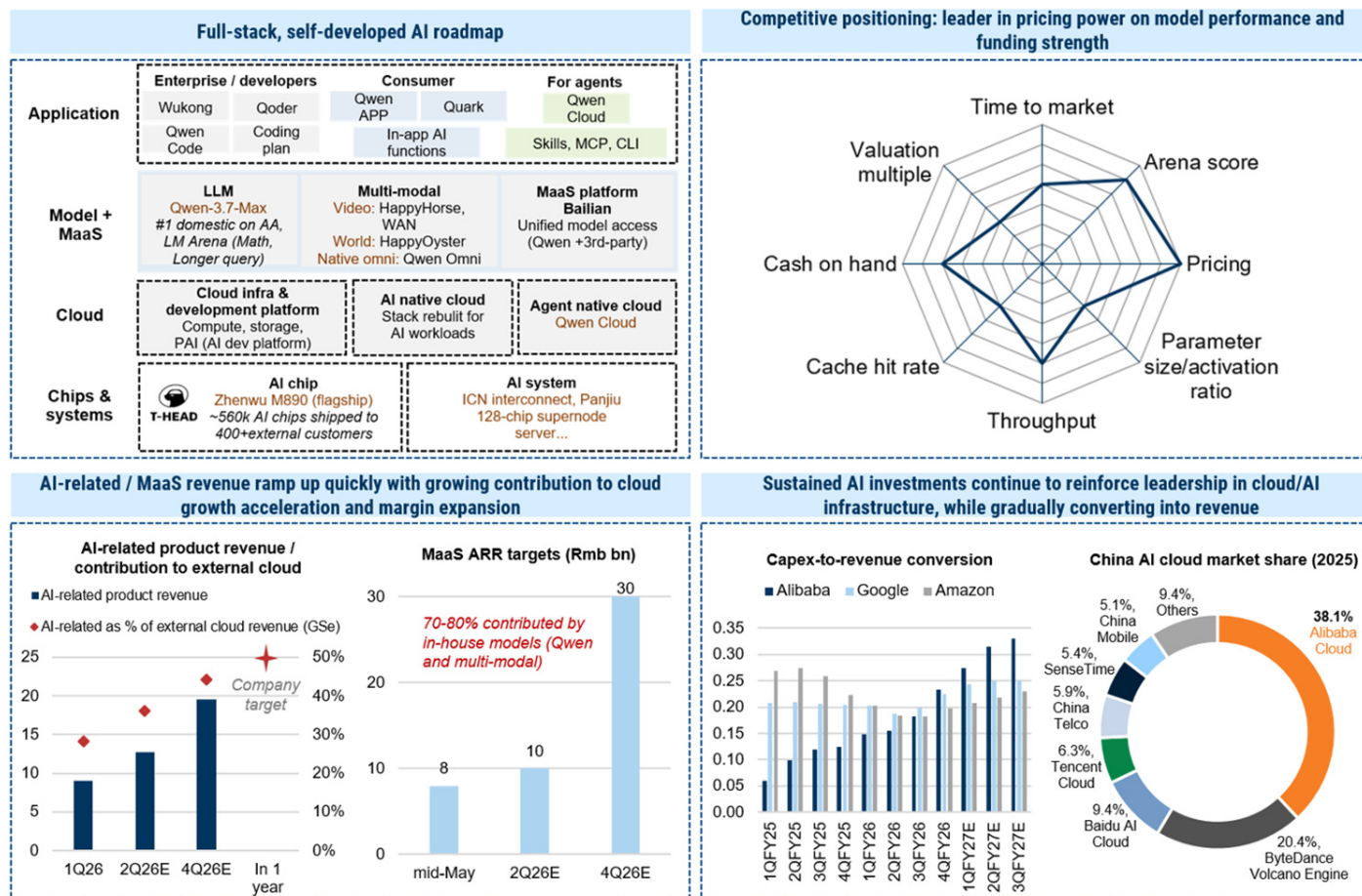
chain risk.

China's key AI players at a glance

Mega-caps

Exhibit 39: Alibaba

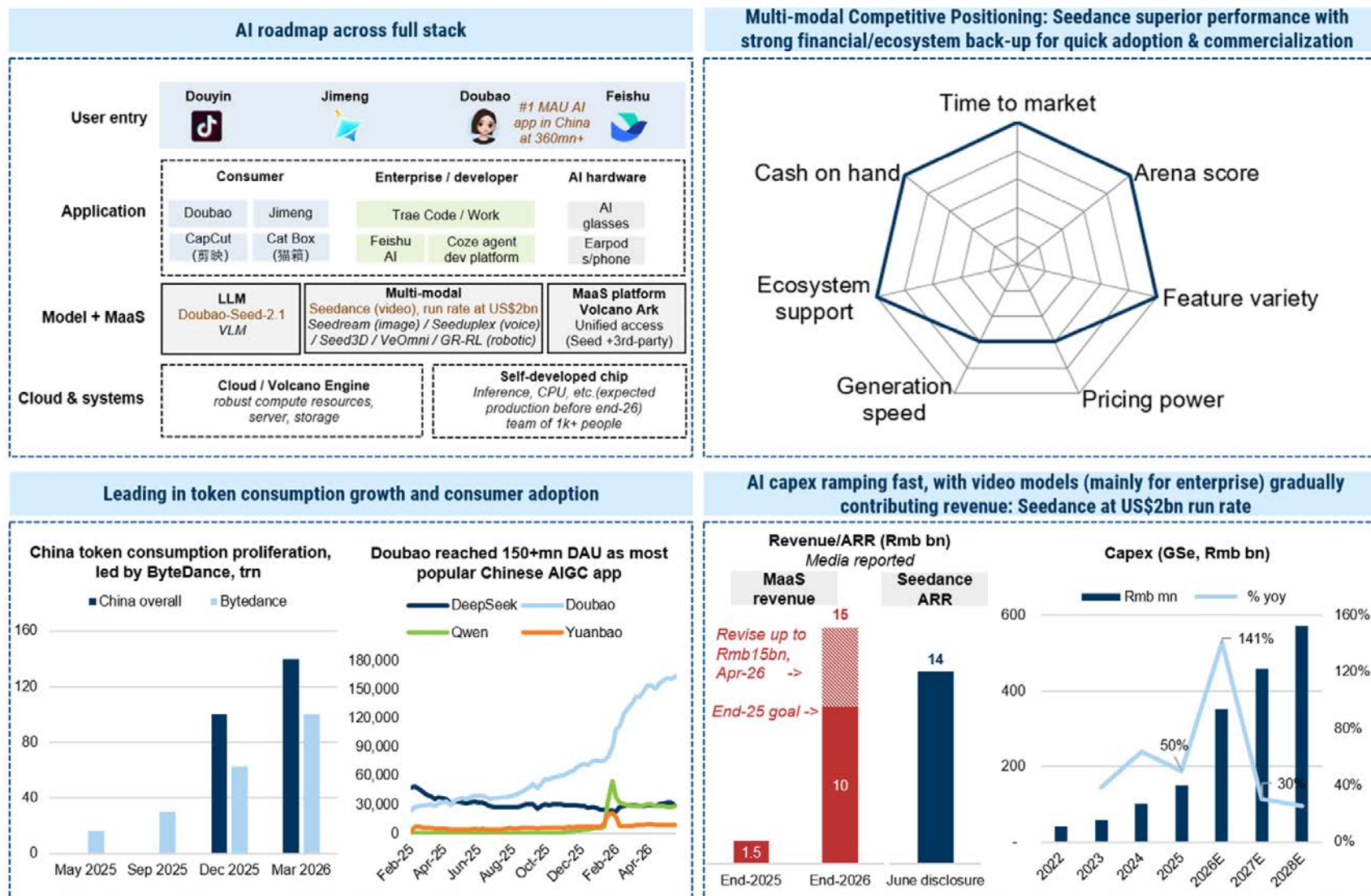
Alibaba: Full-stack AI player with multi-year AI/cloud upside, leading position of most areas



Source: Company data, Goldman Sachs Global Investment Research, Omdia

Exhibit 40: ByteDance

ByteDance: Deep resources for broad AI exploration with strong consumer base and product portfolio

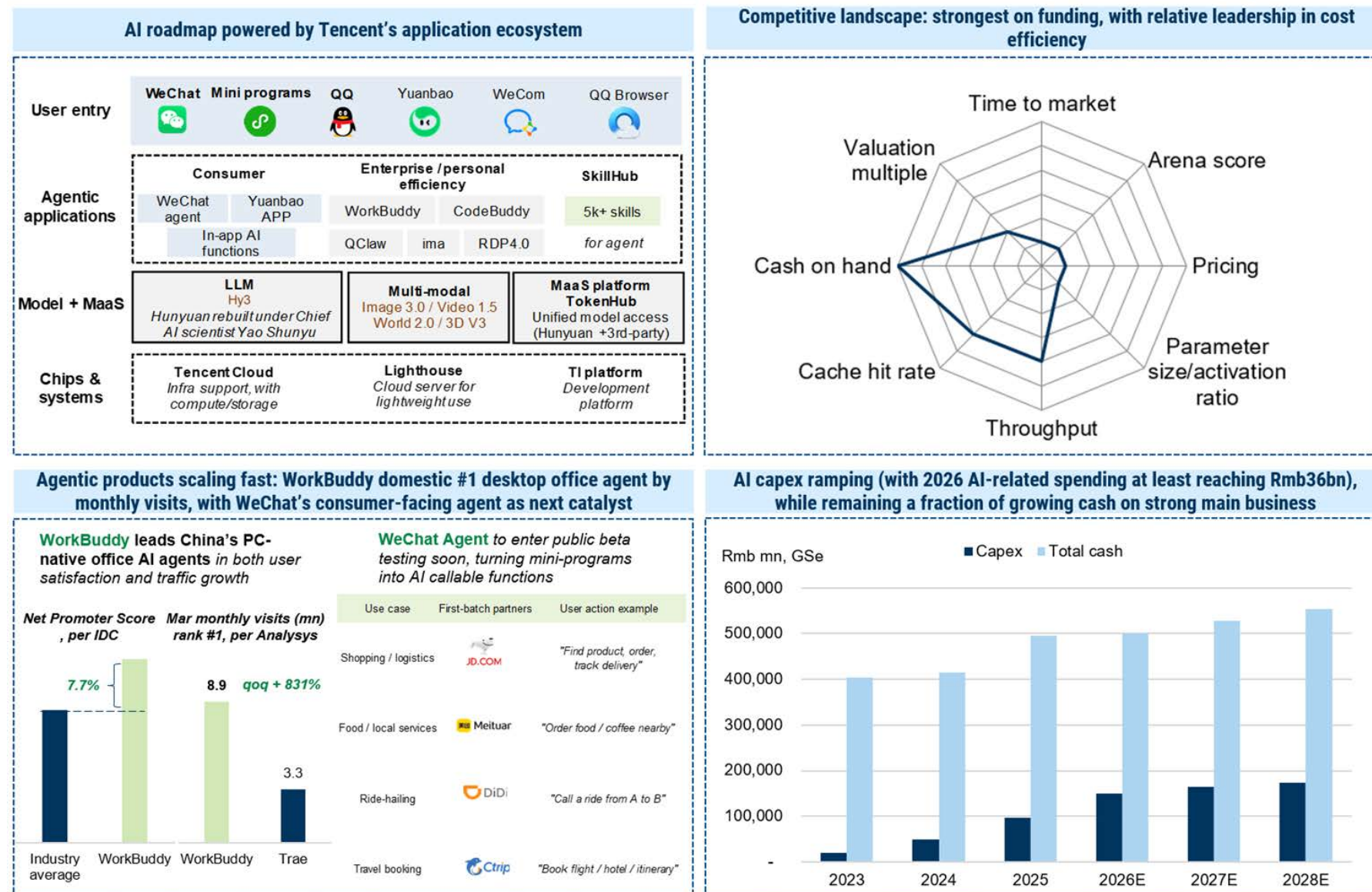


ByteDance's (Not Covered) competitive positioning metrics are calculated by analyzing the industry and companies that we cover, and then extrapolating them to ByteDance. N/A for the valuation multiple in the competitive positioning chart

Source: Company data, Goldman Sachs Global Investment Research, LatePost, QuestMobile

Exhibit 41: Tencent

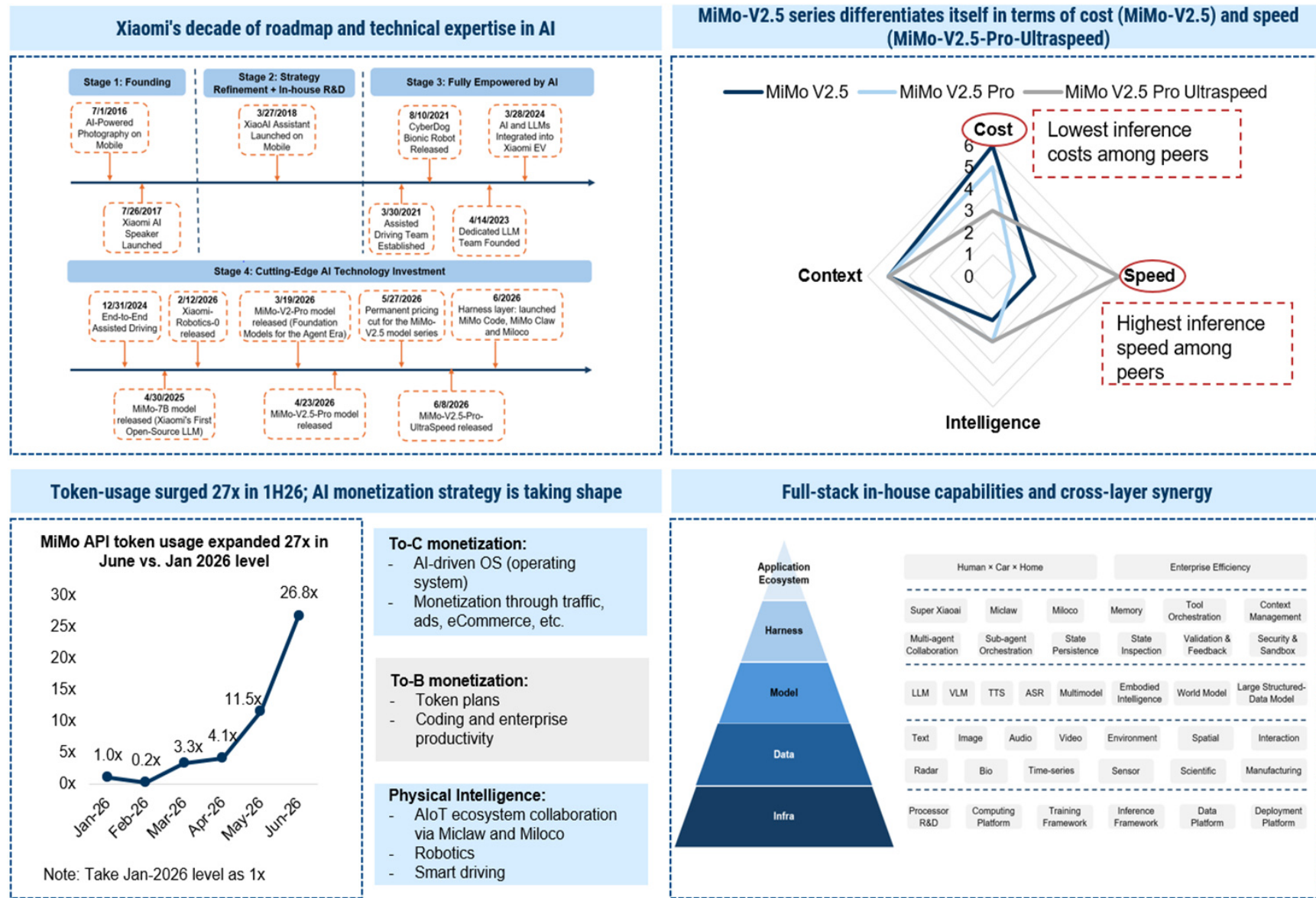
Tencent: Strong contender after AI strategic revamp; potential in Weixin AI agent



Source: Company data, Goldman Sachs Global Investment Research, Analysys, IDC

Exhibit 42: Xiaomi

Xiaomi: Pushes efficiency frontiers of LLMs; Rich applications scenarios in to-C/to-B and physical intelligence

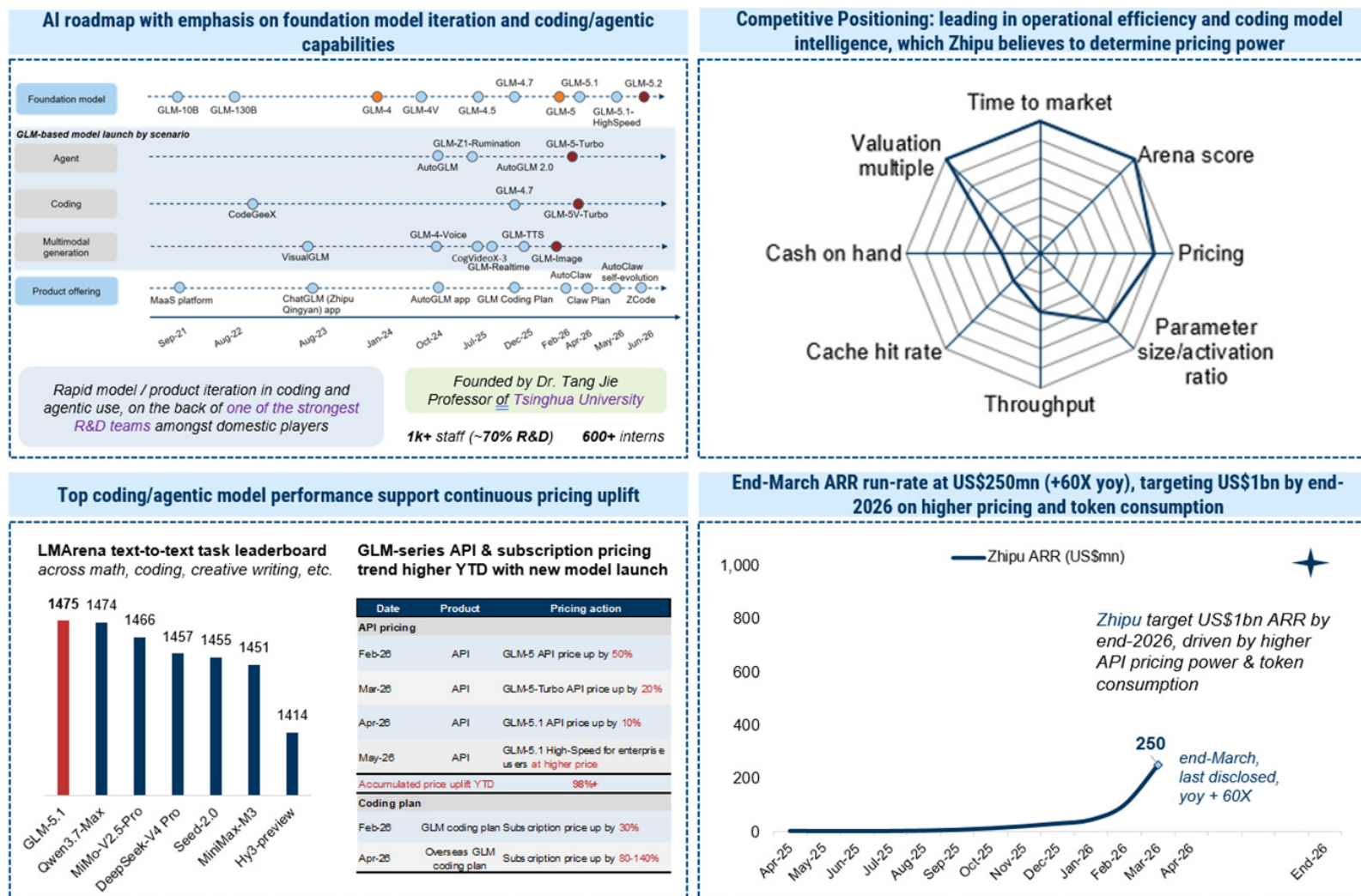


Source: Company data, Goldman Sachs Global Investment Research

Key independent AI players

Exhibit 43: Zhipu

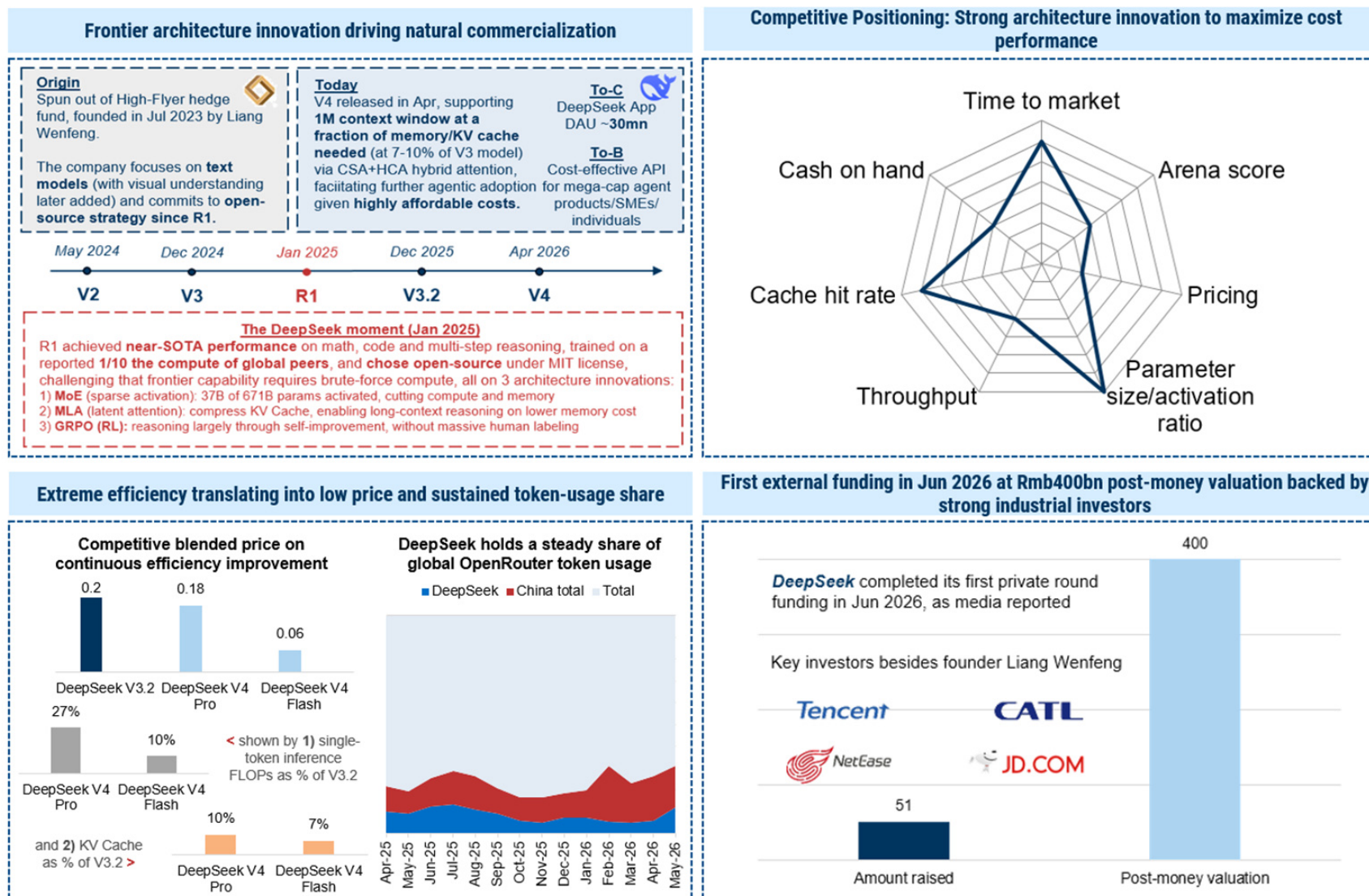
Zhipu: Leading model capabilities in coding/agentic scenarios



Source: Company data, Goldman Sachs Global Investment Research, LMarena

Exhibit 44: DeepSeek

DeepSeek: Extreme cost efficiency to sustain scalable token usage on competitive pricing

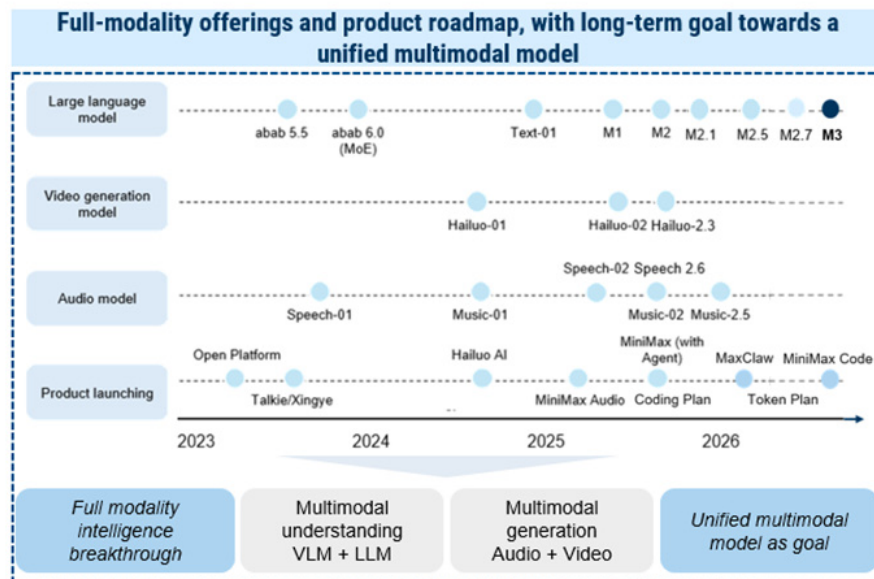


DeepSeek's (Not Covered) competitive positioning metrics are calculated by analyzing the industry and companies that we cover, and then extrapolating them to DeepSeek. DeepSeek's latest funding round based on Reuters.

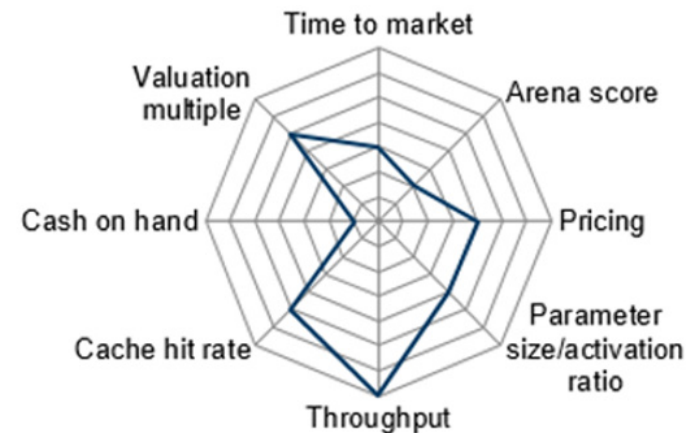
Source: Company data, Goldman Sachs Global Investment Research, OpenRouter, Reuters

Exhibit 45: MiniMax

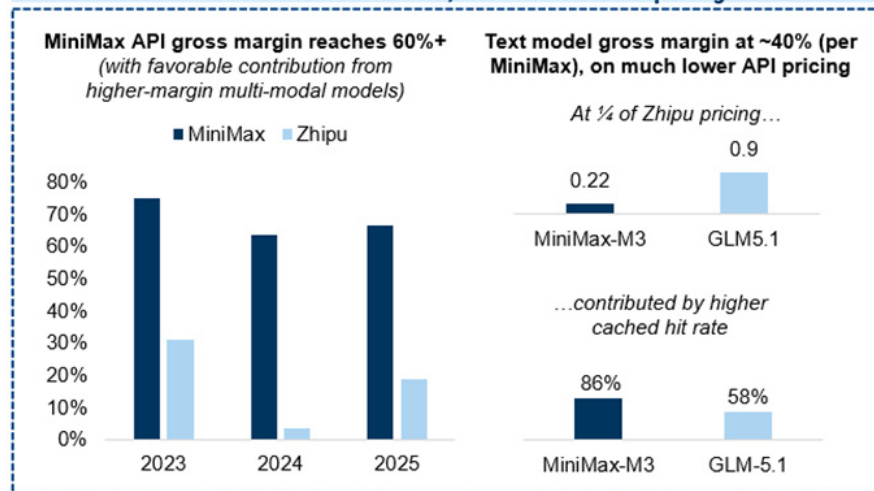
MiniMax: Unique full-modal capabilities among independent players; cost-efficient computing architectures



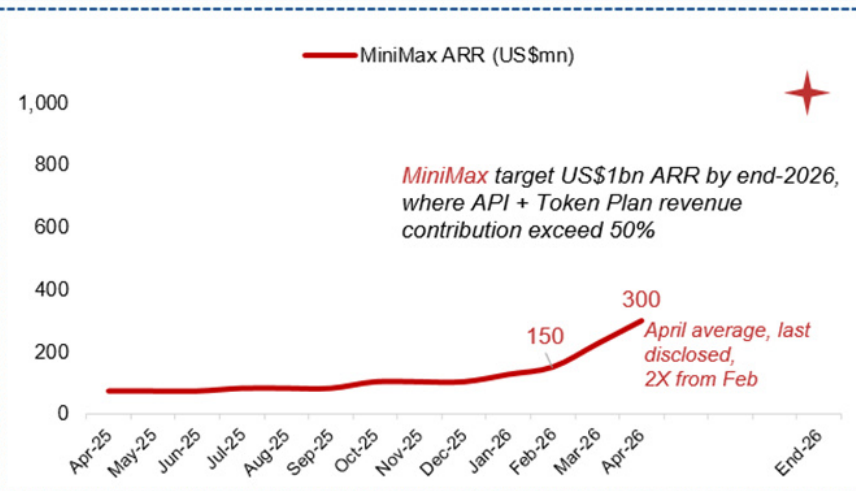
Competitive positioning: strong on multi-modality offerings and cost efficiency among independent players



API gross margin above 60%, with higher-margin multi-modal contribution and ~40% on text models alone, at much lower API pricing



April ARR run rate at US\$300mn (2X from Feb), targeting US\$1bn by end-2026 with API + Token Plan contribution above 50%



Source: Company data, Goldman Sachs Global Investment Research

Price Target Methodology & Risks

Price Target Risks and Methodology – Alibaba Group

Valuation methodology: We are Buy rated on BABA/9988.HK, with SOTP-based 12-month target prices of US\$186/HK\$180 per ADS/share (FY27E-based).

Key risks: (1) Lower-than-expected GMV growth due to macro/competition; (2) Slower-than-expected monetization in China retail; (3) Weaker-than-expected execution in key strategic investments; and (4) Cloud revenue growth deceleration.

Price Target Risks and Methodology – Tencent Holdings

Valuation methodology: We are Buy rated on Tencent (0700.HK), with a 12-m SOTP-based TP of **HK\$700**.

Key risks: More intense industry competition in performance-based advertising, unexpected delays in game launches/Banhao approvals, moderated growth in FinTech and Cloud businesses, slower-than-expected AI development progress, reinvestment risk.

Price Target Risks and Methodology – Xiaomi Corp.

We are Buy rated on Xiaomi with a 12-month target price of HK\$40, based on a SOTP valuation methodology: 1) 16x target 12m-fwd EV/NOPAT on Xiaomi core; 2) DCF-based valuation for Xiaomi EV of US\$42bn (12% WACC and 3% TGR); 3) a 10% holdco discount. Key downside risks: 1) More intense competition and weaker market share gains within the global smartphone industry. 2) Higher GPM pressure on the smartphone/EV business. 3) Below-expected execution on Xiaomi brand premiumization and the EV business. 4) Intensifying geopolitical risks and regulatory uncertainties. 5) Softer macro environment and weaker smartphone/IoT demand. 6) FX fluctuation.

Price Target Risks and Methodology – MiniMax Group

We are Buy rated on MiniMax with a 12-month target price of HK\$860. Our valuation is based on DCF valuation, with 12% WACC and 2% terminal growth rate.

Key risks: Weaker-than-expected model performance amid the competition in the global foundation model industry; Slower-than-expected path to profit visibility; Weaker-than-expected commercialization capability; risks around IP/content generation; cash burn/self-funding capability; risks around geopolitics under an intensified tech race between the US and China.

Price Target Risks and Methodology - Kuaishou Technology

Valuation methodology: We are Buy rated on Kuaishou with a 12m target price of **HK\$68**, based on P/E valuation (target P/E multiple of 15X applied to our avg. 2026E-27E EPS).

Key downside risks: 1) Slower-than-expected ad budget recovery; 2) Weaker-than-expected Kling monetization; 3) Slower-than-expected growth of its aggregate user engagement base (DAU x time spent per DAU); 4) Lower-than-expected profitability; 5) Weaker-than-expected AI progress.

Disclosure Appendix

Reg AC

We, Ronald Keung, CFA, Eric Sheridan, Allen Chang, Lincoln Kong, CFA, Timothy Zhao, Damian Xie, Iris Xiao, Steve Qiu, Luqing Zhou and Eunice Liu, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Ronald Keung, CFA Goldman Sachs (Asia) L.L.C., Eric Sheridan Goldman Sachs & Co. LLC, Allen Chang Goldman Sachs (Asia) L.L.C., Lincoln Kong, CFA Goldman Sachs (Asia) L.L.C., Timothy Zhao Goldman Sachs (Asia) L.L.C., Damian Xie Goldman Sachs (Asia) L.L.C., Iris Xiao Goldman Sachs (Asia) L.L.C., Steve Qiu Goldman Sachs (Asia) L.L.C., Luqing Zhou Goldman Sachs (Asia) L.L.C., Eunice Liu Goldman Sachs (Asia) L.L.C..

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Logo disclosure

Third party brands used in this presentation are the property of their respective owners, and are used here for informational purposes only. The use of such brands should not be viewed as an endorsement, affiliation or sponsorship by or for Goldman Sachs or any of its products/services.

Financial Advisory Disclosure

Goldman Sachs and/or one of its affiliates is acting as a financial advisor in connection with an announced strategic matter involving the following company or one of its affiliates: Baidu, Inc.

Rating and pricing information

Kuaishou Technology (Buy, HK\$40.46) and MiniMax Group (Buy, HK\$323.80)

The rating(s) for Alibaba Group (ADR), Alibaba Group (H), Knowledge Atlas Technology, MiniMax Group and Tencent Holdings is/are relative to the other companies in its/their coverage universe: Alibaba Group (ADR), Alibaba Group (H), DiDi Global Inc., Full Truck Alliance Co., J&T Global Express Ltd., JD Logistics, JD.com Inc. (ADR), JD.com Inc. (H), Jingdong Industrials Inc., Kerry Logistics Network

Ltd., Knowledge Atlas Technology, Meituan, MiniMax Group, PDD Holdings, S.F. Holding (A), S.F. Holding (H), STO Express, Sinotrans Ltd. (A), Sinotrans Ltd. (H), Tencent Holdings, Vipshop Holdings, YTO Express Group, Yunda Holding, ZTO Express (Cayman) Inc. (ADR), ZTO Express (Cayman) Inc. (H)

The rating(s) for Kuaishou Technology is/are relative to the other companies in its/their coverage universe: 37 Interactive Entertainment, Alibaba Health, Bilibili Inc. (ADR), Bilibili Inc. (H), Damai Entertainment Holdings, Focus Media Information Tech, G-Bits, JD Health International, Kuaishou Technology, Mango Excellent Media, NetEase Inc. (ADR), NetEase Inc. (H), Perfect World, Tencent Music Entertainment Group (ADR), Tencent Music Entertainment Group (H), XD Inc., Zhihu Inc. (ADR), Zhihu Inc. (H), iQIYI Inc.

The rating(s) for Xiaomi Corp. is/are relative to the other companies in its/their coverage universe: Autohome Inc. (ADR), Autohome Inc. (H), Beijing Sinnet Technology Co Ltd., East Buy, GDS Holdings (ADR), GDS Holdings (H), KE Holdings (ADR), KE Holdings (H), Kanzhun Ltd. (ADR), Kanzhun Ltd. (H), Kingsoft Cloud, Ming Yuan Cloud, New Oriental Education & Technology (ADR), New Oriental Education & Technology (H), Offcn Education Technology, Range Intelligent, SUNeVision Holdings, Shanghai Athub, TAL Education Group, Tuohi Car Inc., Tuya, VNET Group, Weibo Corp. (ADR), Weibo Corp. (H), Weimob, Xiaomi Corp.

Company-specific regulatory disclosures

Compendium report: please see disclosures at <https://www.gs.com/research/hedge.html>. Disclosures applicable to the companies included in this compendium can be found in the latest relevant published research

Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	60%	45%

As of April 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,074 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Price target and rating history chart(s)

Compendium report: please see disclosures at <https://www.gs.com/research/hedge.html>. Disclosures applicable to the companies included in this compendium can be found in the latest relevant published research

Target price history table(s)

Alibaba Group (ADR) (BABA)

Date of report	Target price (\$)	Closing price (\$)
27-Feb-26	186.00	144.11
25-Nov-25	197.00	157.01
12-Oct-25	205.00	159.01
16-Sep-25	179.00	162.21
31-Aug-25	163.00	135.00
15-Jul-25	147.00	116.97
02-Jul-25	150.00	110.71
21-Apr-25	159.00	110.15
20-Feb-25	160.00	135.97
20-Jan-25	117.00	85.12
29-Sep-24	134.00	107.33
15-Aug-24	108.00	79.54
16-Jul-24	107.00	78.38
15-May-24	106.00	80.99
11-Apr-24	102.00	74.85

Tencent Holdings (0700.HK)

Date of report	Target price (HK\$)	Closing price (HK\$)
18-Mar-26	700.00	550.50
18-Jan-26	752.00	617.50
12-Oct-25	770.00	651.50
13-Aug-25	701.00	586.00
05-Aug-25	658.00	559.00
14-May-25	595.00	521.00
19-Mar-25	590.00	540.00
20-Jan-25	534.00	389.80
13-Nov-24	542.00	403.80
29-Sep-24	521.00	437.80
14-Aug-24	464.00	373.80
12-Jun-24	477.00	370.80
15-May-24	466.00	381.80
11-Apr-24	408.00	315.00
20-Mar-24	413.00	288.80

Alibaba Group (ADR) (BABA)

Date of report	Target price (\$)	Closing price (\$)
07-Feb-24	105.00	73.64
15-Jan-24	121.00	71.84
09-Oct-23	134.00	84.85
10-Aug-23	138.00	99.21
16-Jul-23	133.00	94.56

Tencent Holdings (0700.HK)

Date of report	Target price (HK\$)	Closing price (HK\$)
15-Jan-24	426.00	289.40
15-Nov-23	435.00	322.60
11-Oct-23	423.00	311.00
16-Aug-23	431.00	328.80
07-Aug-23	428.00	341.20

Xiaomi Corp. (1810.HK)

Date of report	Target price (HK\$)	Closing price (HK\$)
26-May-26	40.00	29.76
09-Mar-26	41.00	33.68
23-Jan-26	47.50	36.24
18-Nov-25	53.50	40.78
01-Nov-25	56.50	43.20
25-Sep-25	66.00	59.45
19-Aug-25	65.00	52.40
26-Jun-25	69.00	56.90
27-May-25	65.00	51.55
18-May-25	62.00	51.00
16-Apr-25	59.00	41.25
18-Mar-25	63.00	57.65
20-Feb-25	58.00	49.15
14-Jan-25	38.00	33.75
18-Nov-24	33.30	28.80
29-Oct-24	30.70	25.85
15-Oct-24	27.80	23.00
03-Oct-24	27.50	24.05
21-Aug-24	24.70	17.52
21-Jul-24	23.20	16.52
24-May-24	23.00	18.30
12-May-24	22.60	19.40
01-Apr-24	20.00	14.94
19-Mar-24	18.90	14.86
06-Feb-24	18.10	12.90
20-Nov-23	18.70	16.18
18-Oct-23	16.30	13.18
31-Aug-23	14.90	12.36
09-Aug-23	14.70	12.14

Kuaishou Technology (1024.HK)

Date of report	Target price (HK\$)	Closing price (HK\$)
29-Jun-26	68.00	41.00
27-May-26	70.00	45.28
14-May-26	74.00	50.85
25-Mar-26	75.00	53.05
18-Jan-26	87.00	78.35
31-Aug-25	83.00	75.35
21-Aug-25	77.00	71.75
17-Jun-25	68.00	59.40
27-May-25	63.00	48.75
25-Mar-25	62.00	56.80
19-Jan-25	55.00	40.75
20-Nov-24	63.00	52.55
20-Aug-24	67.00	44.40
08-Aug-24	70.00	43.85
22-May-24	80.00	58.15
20-Mar-24	75.00	52.25
21-Jan-24	82.00	41.55
21-Nov-23	88.00	58.50
19-Oct-23	87.00	56.10
22-Aug-23	91.00	62.50

Alibaba Group (H) (9988.HK)

Date of report	Target price (HK\$)	Closing price (HK\$)
27-Feb-26	180.00	142.90
25-Nov-25	192.00	157.80

MiniMax Group (0100.HK)

Date of report	Target price (HK\$)	Closing price (HK\$)
16-Jun-26	860.00	416.60
03-Mar-26	1,000.00	821.00

Alibaba Group (H) (9988.HK)

Date of report	Target price (HK\$)	Closing price (HK\$)
12-Oct-25	199.00	165.40
16-Sep-25	174.00	153.50
31-Aug-25	158.00	115.70
15-Jul-25	142.00	113.50
02-Jul-25	146.00	109.40
21-Apr-25	154.00	108.70
20-Feb-25	156.00	120.90
20-Jan-25	114.00	84.55
29-Sep-24	131.00	102.50
15-Aug-24	105.00	76.40
16-Jul-24	104.00	75.50
15-May-24	103.00	82.65
11-Apr-24	99.00	74.20
07-Feb-24	102.00	74.90
15-Jan-24	118.00	70.00
09-Oct-23	131.00	82.10
10-Aug-23	134.00	94.30
16-Jul-23	130.00	92.95

MiniMax Group (0100.HK)

Date of report	Target price (HK\$)	Closing price (HK\$)
22-Feb-26	1,018.00	970.00

Price targets shown in table(s) are unadjusted for corporate actions.

Regulatory disclosures**Disclosures required by United States laws and regulations**

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research

may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: <https://www.goldmansachs.com/worldwide/india/research-analyst>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for “professional investors” within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither “registered banks” nor “deposit takers” (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for “wholesale clients” (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom’s departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock’s total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region’s Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts’ investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at <https://www.gs.com/research/hedge.html>.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs’ involvement in a transaction, and in certain other circumstances.

Early-Stage Biotech (ES). An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in

Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International (“GSI”), authorised by the Prudential Regulation Authority (“PRA”) and regulated by the Financial Conduct Authority (“FCA”) and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE (“GSBE”) is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst’s judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst’s published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst’s fundamental equity rating for such stocks, which rating reflects a stock’s return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts’ fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst’s fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.

MINDCRAFT: OUR THEMATIC DEEP DIVES

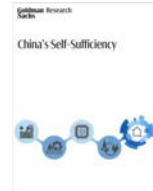
Artificial Intelligence



Humanoid Robots



China Self-Sufficiency



Space



Navigating China Internet



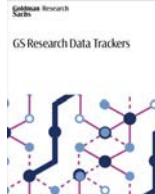
The Ecosystem of EV



Ecosystems



GS Research Data Trackers



Autonomous driving



China Biopharma



China Consumer



CHIPS Act Impact



Investing in India



Resurgent Japan



China Grid Tech



China Property



Supply Chain Shifts



Trade and tariffs



Market Concentration



Top of Mind



Carbonomics



Market Cycles



Tracking the Consumer



Korea Value in Action



Cybersecurity and Defense



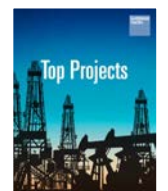
Computing Advances



Magnificent 7



Top Projects



Obesity



Geopolitics



Understanding China's Statistics



ESG

